An Evaluation of Alternative Matching Techniques for Use in Comparative Interrupted Time Series Analyses:

An Application to Elementary Education

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by

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Abstract

Comparative interrupted time series (CITS) models attempt to identify the causal effect of a policy change on a treated group, such as a school, by determining the difference in outcome before and after the policy change, using the pre/post difference in the comparison group as the counterfactual. The motivation for the current study stems from application of a CITS to estimate the relationship between student achievement and implementation of a magnet school program which requires selecting as comparisons traditional non-magnet schools whose changes in achievement reflect the counterfactual. Unfortunately, no consensus exists on how to appropriately select a comparison group. To this end, we conducted a Monte Carlo simulation in which we compared four methods of selecting a comparison group, including all traditional public schools in the district, regression-based matching, propensity score matching, and a synthetic control group method. Based on a sample of 24 school districts, we performed a falsification exercise by randomly assigning one elementary school per district to hypothetical treatment status, selecting a comparison group, and testing for a statistically significant effect of the hypothetical treatment. This process was repeated 1,000 times. In most cases, the method that came closest to rejecting the true hypothesis of no effect at the conventional five percent significance level is the method that uses all other traditional public elementary schools in the district as the comparison group. We also provide separate analyses for varying district sizes and number of time periods of data available. These results should help future education researchers using CITS models think about which method to use in selecting an appropriate comparison group.

Executive Summary

The paper evaluates the performance of alternative methods for identifying groups of comparison schools for use in the National Evaluation of Magnet Schools (NEMS), a study of student achievement in elementary schools that used federal Magnet Schools Assistance Program (MSAP) funds to convert from traditional to magnet schools. The evaluation uses a comparative interrupted time series (CITS) design in which mean ELA and math scores of magnet schools are compared to mean scores for similar non-magnet schools chosen from the same district to serve as comparisons. Because this method is widely used in the social sciences, the investigation may be useful to researchers in other areas. [1]

In the context of the NEMS study, the pool of schools from which a comparison group can be drawn varies considerably among the districts. During the NEMS feasibility phase, a preliminary set of comparison schools was identified by hand, taking account of test score means in the pre-magnet years, one-year changes in school-level ELA and math scores, and demographic composition data. This paper evaluates several systematic methods for selecting comparison groups, which are less subjective than hand-matching methods used in the feasibility phase and in other previous studies. Although a variety of approaches has been described in the literature, it is as yet unclear which methods would be most appropriate for CITS models in general and for the schools and districts in the NEMS in particular.

Using a Monte Carlo simulation approach, this study compared four general methods of identifying comparison groups:

1. Use all traditional public schools in the district as the comparison group
2. Use regression models of the dependent variable (test scores) to identify the "best" matching comparison schools as those with predicted changes in post-magnet conversion test scores that are closest to that of the magnet school
3. Use a propensity score model to match based on the predicted probability that a school becomes an MSAP-funded magnet school
4. Use a weighted combination of non-magnet schools as the comparison group, where weights assigned to each non-magnet school are generated by a procedure introduced by Abadie and Gardeazabal (2003), which minimizes observable differences between the treatment and comparison schools and used to form a "synthetic control" comparison group.

The study used ELA and math test score data for school years 2001-02 through 2006-07 from the National Longitudinal School Level State Assessment Score Database (NLSLSASD) and school-level demographic data from the National Center for Education Statistics (NCES) *Common Core of Data* for those years. The data were drawn from selected districts in three states for which state tests did not change over the six-year period (California, South Carolina, and Florida). The simulation used data from the elementary schools in each of the selected districts that were not charter schools and that did not have magnet programs during these years.

[1] Economists typically refer to this approach as difference-in-difference models.

To evaluate the performance of each matching approach, we conducted a falsification exercise. We began by designating a traditional public school in each of 24 districts as a hypothetical magnet school based on a stratified random sampling strategy in which lower performing schools on state achievement tests had a greater probability of being selected. Next we used the given matching approach to identify a comparison group. After pooling schools across all districts, we used regression analysis to calculate a hypothetical magnet school effect that measured the average difference in test scores between the randomly assigned "magnet" and comparison schools during the treatment period. This estimated effect excludes any persistent pre-interruption differences among schools and also excludes any changes that can be accounted for by changes in the other explanatory variables. A Monte Carlo simulation was performed where the process was replicated 1,000 times. The distribution of estimated effects across the 1,000 replications was then used to gauge the quality of the matches. The matching methods were compared with a view to identifying the method which was closest to rejecting the null of no magnet/comparison school difference at the expected level of five percent. In the case that two or more methods tied for best, we then chose the method that had a mean "effect" closest to zero, and after that the method that produced effects with the smallest standard deviation across replications.

The study evaluated the four approaches mentioned above while varying the following conditions:

1. District size (i.e., districts with few, a moderate number, or many potential comparison schools)
2. The number of years of data on test scores and school demographic characteristics prior to the hypothetical magnet school adoption used to match schools (two versus three years) and similar variations in the number of years of test scores after magnet programs had been "adopted."
3. Matching using test scores only versus matching that also included school demographic characteristics in the matching procedure.
4. Identifying comparison schools based on a composite of math and ELA scores versus identifying separate sets of comparison schools for analyzing the schools' performance on math and ELA.

In the simulations, the method that rejected the null hypothesis of a zero magnet effect the closest to five percent of the time was the one that used all non-magnet and non-charter schools in the same district as the comparison group. For instance, when we used three years of pre-treatment and three years of post-treatment data, using all 24 districts of various sizes, for English Language Arts, the method that used all traditional schools as the comparison group rejected the null 6.5 percent of the time, compared to 7.3 percent, 7.8 percent, and 7.2 percent for the regression-based, propensity score and synthetic control approaches. For math models, the corresponding percentages were 6.9, 7.2, 7.6 and 11.6 percent, respectively. Thus, using all potential comparison schools "over-rejected" the least, compared to the expected rejection rate of five percent. On the full sample of 24 districts, this general finding was not sensitive to the number of years of data available pre- and post-treatment.

When we repeated the overall analysis for small, medium and large school districts, we found that the method that used all traditional schools as a comparison group continued to excel in medium and large districts. But in small districts propensity score matching led to the lowest over-rejection rates in models of math scores or the average of math and ELA scores, while the regression-based matching method over-rejected the least of all tested methods when ELA scores were modeled.

It may seem surprising that the method that used all the traditional schools in a district, which represented all the potential comparison schools, tended to over-reject to a lesser degree than the other methods, given that each of the latter explicitly attempts to match the magnet school to comparison schools. However, the key requirement for the comparative interrupted time series model to produce an unbiased estimate is that in the treatment period the counterfactual pattern of test scores for the treated school, had it not been treated, should be parallel to the pattern of test scores observed for the comparison schools. Each of the three methods that explicitly attempts to match does so based on variables in the pre-treatment period only. They will produce biased results if the comparison schools selected based on the pre-treatment data do not track the counterfactual post-treatment pattern of test scores for the treated school. Thus, matching based on pre-treatment values of variables does not guarantee a good match in the post-treatment period.

The "all traditional schools" method that uses all potential comparison schools may reduce the chances of bad matches simply because unusual year-to-year fluctuations in the outcome variable among comparison schools in the treatment period tend to average out as the number of comparison schools rises. Put differently, the variance of the average post-treatment patterns in the outcome variable will fall as the number of comparison schools rises.

In other findings, the method that uses regression-based matching performs slightly better, especially in large districts, when we do not condition on student demographics relative to models in which we do condition on these variables. A likely explanation, that also supports the earlier result that using all comparison schools typically over-rejects less than the regression-based method, is that by conditioning on demographics as well as lagged test scores, collinearity leads to poor matching in some cases. Conversely, the synthetic control method over-rejects slightly more often when matching on demographics is dropped.

For the regression-based matching and synthetic control methods, we also compared outcomes when we matched based on the average of ELA and math scores, rather than on the specific math or ELA test scores that served as outcomes in the CITS regression. This approach may be preferable if one wanted to have exactly the same comparison group when modeling the effect of treatment on both math and ELA scores. For the regression-based matching method, using the average of these two test scores led to rejecting closer to five percent of the time than when we matched using just one subject test, perhaps due to the greater precision afforded by averaging. No clear pattern emerged when we used the average of math and ELA scores in the synthetic control approach.

For researchers working in the context of evaluating school reforms, the main lesson from these simulations is that we may need to reconsider the practice of selecting subsamples of comparison schools in CITS models. Indeed, researchers may do more harm than good when estimating CITS models by selecting a subsample of the untreated schools within the district as the comparison group. Quite often, using all potential comparison schools in the given district may prove more reliable. The extent to which this is true depends, however, on the mix of the sizes of the districts in terms of the number of potential comparison schools.

Introduction and Motivation

Use of Comparison Groups in Quasi-Experimental Studies

Comparative interrupted time series (CITS) models use changes in outcomes over time to estimate the effects of program and policy changes on these outcomes.[2] Researchers observe an outcome before and after a policy change for a treated group, from which they can estimate the before/after difference. But pre-existing trends could account for some of this difference, so the researchers calculate the analogous before/after difference in the outcome over the same period for a comparison group. By taking the difference in these two differences, researchers hope to approximate the causal effect of the policy intervention, net of pre-existing trends.

Social scientists have implemented comparative interrupted time series (CITS) models in a variety of contexts. Shadish, Cook and Campbell refer to this type of model as an interrupted time series design with a nonequivalent no-treatment control group and provide examples from a wide range of subject areas within the social sciences (pp. 182-184, 2002). Economists have also widely adopted this approach, as have educational researchers in evaluating the effects of various education reforms. (See for instance the analysis of Accelerated Schools by Bloom et al., 2001.)

The success of the CITS method depends entirely on how well the pre/post difference in outcomes for the comparison group matches the pre/post difference that would have happened to the treated group had the reform never taken place. Because one can never observe the outcome for the treatment group in an untreated state, often referred to as the "counterfactual", comparison groups are typically selected based on their similarity to the treatment group in terms of pre-existing outcomes. The implicit assumption is that these pre-existing outcomes are adequate predictors of the expected post-program outcome in the absence of the program or policy change.

In spite of the importance of treatment and comparison groups being similar with respect to pre-existing patterns in the outcome, relatively little work has been done to compare alternative methods of selecting an appropriate comparison group. This study contributes to the literature by determining the preferred method for identifying comparison schools for use in a CITS framework. It was motivated by a particular study the authors are performing, the National Evaluation of Magnet Schools (NEMS), which seeks to evaluate the achievement outcomes of magnet schools that received a federal grant versus those that did not adopt a magnet program. Because many intervention studies focus on public elementary schools, we believe that the results will be of direct relevance to evaluations of other education interventions, and more broadly in other domains of study.

For the NEMS, investigators will examine the relationship between elementary school adoption of a whole school magnet program funded by the federal Magnet Schools Assistance Program (MSAP) and achievement of resident students in each of the years following its adoption.[3] Resident students

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[2] These models are commonly referred to by economists as difference-in-differences or "diff-in-diff" models.
[3] For the analysis plan, see Betts, Levin, Christenson and Eaton (2008). The design document that led up to this

are defined as those students who live in the local attendance zone of the school they are attending, and therefore are automatically admitted because it is their neighborhood school. The relationship between magnet school conversion and student achievement will be examined using a CITS model in which the average achievement of students in magnet and non-magnet schools is compared for cohorts prior to and after the date of conversion to magnet school status. The conversion of a traditional school to magnet status constitutes an "interruption" that distinguishes the periods before and after the reform is in effect. If the average post-interruption achievement score of resident students attending magnet (treatment) schools is significantly higher than that of their non-magnet (control) school counterparts *relative* to what would be expected based on pre-interruption differences in achievement, we may conclude that the conversion was successful in increasing student achievement.[4,5]

Identifying Comparison Groups Based on Pre-Interruption Achievement

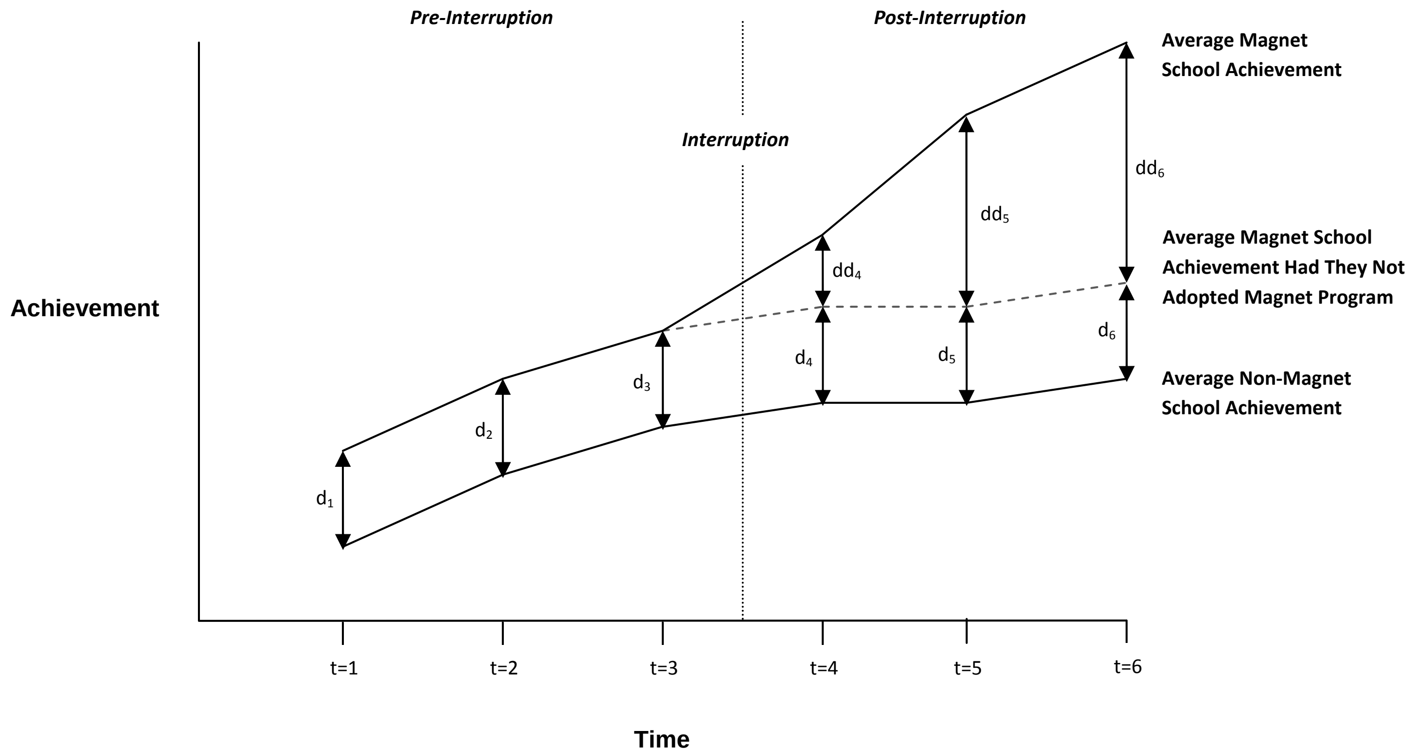
Figure 1 shows the hypothetical average achievement at six points in time for a group of schools that adopted a magnet school program between time t=3 and t=4 ("Average Magnet School Achievement") and a group that did not ("Average Non-Magnet School Achievement"). These hypothetical profiles represent achievement trajectories over time for the groups of treated (magnet) and untreated (non-magnet) comparison schools. In this example there are pre-existing differences in achievement favoring the schools that eventually become magnet schools. While the pre-interruption differences in average achievement between the two groups are constant, these differences increase after adoption of the magnet school programs. This increase in difference indicates that the magnet program is associated with, and may indeed cause, greater student achievement.

examination of magnet schools receiving MSAP funding (Bloom, Doolittle, Garet, Christenson and Eaton, 2004) suggests comparing changes in test scores of resident magnet school students occurring before and after a given school converts to magnet school status with changes observed for two or three comparison schools. The document suggests selecting comparison schools that resembled the magnet school in test-score changes prior to the year in which the magnet school converted. The Bloom et al. (2004) design document provides the specific motivation for the study proposed here.

[4] The term "significant" indicates that the magnitude of the pre/post Intervention difference in the observed magnet/non-magnet school achievement differences (i.e., the "difference-in-differences") is larger than would be expected merely by chance.

[5] Note that taking the difference of the treatment and comparison pre/post differences and taking the pre/post difference of the treatment/comparison group differences are mathematically equivalent representations of the CITS (difference-in-differences) estimator.

Figure 1 –Hypothetical Example of Comparative Interrupted Time Series Analysis (CITS) of Changes in School Achievement Related to Adoption of a Magnet Program



For the CITS approach to work, it is crucial that changes in the achievement outcome for the comparison group after the interruption closely match those that would have occurred in the treatment group, had it not received the policy intervention. To see this in the graphic example above, consider the dotted post-interruption profile for magnet schools that represents the "counterfactual" average achievement for this group under the crucial assumption that *they would have maintained a parallel profile to the non-magnet schools had they not adopted a magnet school program*.[6] In the example above, because the profiles are identical between the two groups (i.e., the magnet and non-magnet series are parallel), the differences are constant at each time period (i.e., $d_1= d_2= d_3= d_4= d_5=d_6$). Let us define this collection of average magnet school achievement measures across the six periods as the "normal" magnet school profile and the distances d_1 through d_6 as "normal" differences (i.e., those magnet/non-magnet differences that would have been observed if the magnet schools did not in fact adopt a program). Deviations from the assumed "normal" magnet school profile in the post-interruption period, as shown in the solid magnet school profile in Figure 1, constitute the CITS estimates of the magnet school program. These estimates are denoted as dd_4, dd_5 and dd_6 in the graph.

[6] In practice, regression analysis is often used to apply the CITS approach so that it is not completely necessary for the pre-treatment trends to be identical between the treated and untreated groups. If a time-varying set of explanatory variables can account for any deviations in pre-treatment trend, the method will remain valid.

Importantly, if the counterfactual treatment group trend is not parallel to the trend for the chosen comparison group as represented in Figure 1, then the CITS estimator will be biased.[7] In reality, it is difficult to identify a group of comparison schools for which the average achievement profiles parallel those of the treatment group schools in the pre-interruption period. What is even more difficult (or even arguably impossible) is to prove that the chosen comparison group exhibits a post-interruption trend that matches what the treatment group would have experienced in the absence of treatment. However, as will be shown below, in practice it may be possible to consistently model year to year changes for the treatment and comparison groups based on observable pre-interruption characteristics.

Although we have illustrated the importance of properly matching treatment and comparison schools, the question remains as to what methodology should be used for such an undertaking. We addressed this challenge by using a simulation analysis to compare the performance of four alternative matching strategies. The performance of each matching strategy was evaluated based on a series of falsification tests. We used actual historical achievement data to run simulations in which we assigned a hypothetical magnet treatment status to a stratified random sample of traditional schools at the mid-point of the six year period, and tested the true null hypothesis that their average post-interruption test score changes should not differ from the changes for comparison schools by more than would be expected by mere chance. We used school-level performance data in ELA and mathematics from the six school years from 2001-02 through 2006-07 for 24 school districts in three states (California, Florida, and South Carolina). These states were selected because each reports student achievement outcome data, measured in a consistent way over the entire six-year period, which are publically available from the National Longitudinal School-Level State Assessment Score Database (NLSLSASD) and state web sites. The falsification exercise excluded all actual magnet and charter schools as part of either the treatment or comparison group. It assumed that the hypothetical intervention occurred between the 2003-04 and 2004-05 school years (analogous to times t= 3 and t=4 in Figure 1), so that 2004-05 denotes the first year of the hypothetical treatment.

Comparison Group Selection Methods

A key methodological question in any CITS analysis is how best to choose the comparison group with which to model the unobserved counterfactual year to year changes for the treatment group. The question here is not only what method to employ to identify appropriate comparison group observations, but also what available information to use in the matching process. That is, if researchers have data for variables other than the outcome of interest, such as student demographics, how should this information be included in the matching process? Moreover, do certain methods perform better in different contexts related to the number of years of available pre/post-interruption data and the size of pool of candidate comparison units? This study investigated the performance of four specific methods to select an appropriate comparison group for a CITS evaluation:

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[7] Appendix B provides a graphical illustration of the bias that occurs when the comparison/treatment group and comparison/counterfactual profiles are not parallel.

- Use All Traditional Public Schools – This approach simply includes all traditional public schools in the comparison group when they are not the school identified as the hypothetical magnet school. By maximizing the size of the comparison group, this approach is also likely to increase the statistical precision of the estimated treatment effect. However, if the assumption of parallel achievement trajectories between the treatment and comparison groups is violated this increase in statistical power may come at the cost of greater bias in the estimates.

- Regression Based Matching – This approach is based on a regression model that estimates pre-interruption achievement for all schools (both the hypothetically treated and untreated). We selected comparisons by choosing those untreated schools whose *predicted change in achievement* most closely matches the predicted achievement change of the hypothetical treatment schools for the pre- to post interruption period (2003-04 to 2004-05). It is important to note that the estimated regression makes use of the achievement outcome and covariate measures observed *before* treatment is introduced (i.e., before the interruption). The predictions were generated from equations estimated using lagged and twice lagged values of achievement levels, incidence of eligibility for federally subsidized free or reduced price lunch, and minority composition percentages [8]

- Propensity Score Matching – This approach matches on the probability of receiving treatment rather than on the predicted outcome of interest –student achievement. We used a probit model to predict the real life incidence of MSAP magnet school status in 2005-2006 for schools in all districts in the three states in this simulation study, and selected comparisons by choosing those untreated (i.e., non-MSAP) schools for which the *predicted probability of receiving treatment* most closely matched that of the treatment (i.e., MSAP magnet) schools. Similar to the regression-based matching method, the estimated regression used in propensity score matching makes use of covariate measures observed *before* treatment is introduced (i.e., before the interruption).[9] The probit model conditioned MSAP magnet school status on current-year percentage of students eligible for f*ree lunch assistance, the percentage African*-American, and the percentage Hispanic, along with values of these three variables in the previous two years.[10]

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[8] We estimated different models where we varied the number of years of pre- and post-interruption data used in order to test the sensitivity of the results to the number of years of available used. The runs employing three years of pre-interruption data were used to model a one-year change in achievement (from t=2 to t=3) based on two years of lagged achievement and other covariates, while those using two years of pre-interruption data modeled the achievement level in the year prior to the interruption (t=3) based on two years of lagged achievement and other covariates. We used forecasts of these variables for the first year of treatment (t=4) for the purposes of choosing comparison schools. Moreover, we experimented with two variants of these models: one based on lagged test scores and student poverty as covariates and a second that conditioned on lagged test scores, student poverty, and student demographic characteristics. Appendix C provides a technical discussion of these models.
[9] The 2005-06 school year was chosen as the hypothetical interruption year for this method because this was the only year in which all of the required data were available for all three states.
[10] The fit from this model was not particularly good, with a pseudo R-squared of 0.09. Only 1 of 352 MSAP funded magnet *school*s had a predicted probability of being a magnet of over 0.5. In cases where we had only two years

- Synthetic Control Method – This approach, pioneered by Abadie and Gardeazabal (2003), creates comparison schools by constructing a weighted average of all untreated observations to form a "synthetic control" that, by construction, comes as close as possible to mimicking what the treated schools would have experienced in the absence of receiving treatment. Weights are calculated based on pre-interruption values of the observable characteristics by formulating a weighted combination of the comparison schools based on the vector of weights that minimizes the sum of squared differences of the observables between the treatment and comparison schools in the periods before treatment begins. One then can evaluate the treatment effect by comparing post-interruption achievement for the treated schools against the weighted-average synthetic control schools.[11]

Appendix E outlines other possible approaches to selecting a comparison group. It also indicates the reasons we did not adopt these alternative methods.

Research Question – Which Method Works Best?

Our central question is: "Which of these methods performs best at matching the test-score achievement of magnet and non-magnet elementary schools?" While we focus on elementary schools because our primary goal is to inform the federal evaluation currently underway of elementary magnet schools funded by the MSAP program, the results may also be relevant for general use of CITS models in education and other policy area studies.

Simulation Overview

We conducted a simulation to provide a comparative evaluation of each of the four aforementioned methods of selecting comparison schools. The approach was to assign a traditional public school to a hypothetical treatment status in years four through six of the data, use each of the four methods to select comparison schools, and then, through repeated replication, test for each approach how much the achievement outcomes in the hypothetical treatment schools deviated from that of the non-treatment group in the post-interruption years (2004-05 through 2006-07).[12] The hypothetical treatment schools were selected using a stratified random sample in which lower performing schools on state achievement tests had a greater probability of being selected. (This is

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of pre-treatment data, we used a probit model that used as explanatory variables current year data plus a single lag of the variables listed above. The fit of this model was slightly worse, with a pseudo R-squared of 0.06.

[11] An in-depth description of the weighting procedure used to create synthetic control units is presented in Appendix D. As with the regression based method, the basic method creates synthetic control units (weighted combinations of the comparison schools) based on all available pre-interruption values of the outcome variable, while the second version also takes into account other student demographic characteristics that might have predictive power. As with the regression-based matching and propensity-score methods, we examined models that reflected either two or three years of available pre-treatment data for the matching.

[12] The first method described above (use all comparison schools) does not in fact select a subset of the control candidates, but rather includes the full pool of non-treatment schools as a comparison group.

consistent with the profile of typical federally funded magnet school before it becomes a magnet school). The potential comparison schools were restricted to the set of elementary schools in the same district, on the grounds that only in this way could one adequately control for unobserved district policy changes. In this Monte Carlo study, the "experiment" was run 1,000 times for each method, providing a distribution of results for each of the four methods tested. Specifically, the simulation followed these steps:[13]

1) Select the sample of traditional public elementary schools in the three study states that have been verified as not having adopted a magnet school program or been a charter school at any point in the six years studied.
2) Within each state, identify the nine districts that rank at the 30th, 40th and so on through the 90th percentiles as well as the 95th percentile of the nationwide distribution of district size, where district size is calculated as the number of elementary schools identified in step 1 as not having been either a magnet or charter school during the period studied.
3) Within each of the districts identified in step 2, using a stratified random sampling design where schools in the lower achievement strata have a higher probability of being sampled, randomly select one school to be assigned to hypothetical magnet school (treatment) status in the post-interruption years (2004-05, 2005-06 and 2006-07).[14]
4) Match the randomly assigned hypothetical magnet schools from 3) to one or more comparison schools within the same district using the four matching methods discussed above.[15]
5) Perform regressions of outcomes from the identified matched groups of hypothetical magnet and comparison schools using the equation detailed in the next section, and save the estimate of the CITS treatment effect coefficient, as well as its corresponding probability value.
6) Repeat steps three, four and five 1,000 times, selecting a different sample of randomly assigned treatment schools for each run.

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[13] We skip several details for now to convey the overall method more clearly. We will provide these details, mostly on how we pick districts, later.

[14] Schools were stratified into within-district tertiles of ELA achievement with the lowest, middle and highest tertiles given weighted sampling probabilities of 50%, 40% and 10%, respectively. This stratification approach was based on where the nationwide population of 2004 and 2007 MSAP-funded magnet schools tended to be located in the distribution of achievement within their districts; about 50% of the MSAP magnet schools in each cohort were in the bottom third of achievement within their district, 40% were in the middle third, and only 10% were in the top third of the within district achievement distribution.

[15] Note that the second method mentioned above (regression-based matching) will result in selecting two comparison (non-magnet) schools that are the closest match to the hypothetical magnet school based on the pair with an average predicted outcome that is closest to that of the randomly selected treatment school. Similarly, our approach to propensity score matching (the third method) is to pick the two potential comparison schools for which the average predicted probability of becoming a magnet school is closest to that of the randomly selected treatment school. In contrast, the first approach described above (use all comparison schools) uses all other schools in the district, while the final method (synthetic control group) creates a synthetic control group out of a variable number of schools in the district.

This procedure yielded a database containing the estimated treatment effect coefficient, and corresponding t-statistics and p-values for the 1,000 estimated models resulting from each of the four matching approaches, respectively. The results were then evaluated in the spirit of a falsification test, where methods that showed abnormally high rates of statistically significant difference in outcomes between the hypothetical magnet and chosen comparison schools were deemed less valid. The simulation used up to six years of data (from 2001-02 through 2006-07) for each school district in our three-state sample (California, Florida and South Carolina).

Outcome Equation

The estimated outcome equation modeled test scores Y_{sdt} for school *s* in district *d* in year *t*. For the regression-based matching and propensity-score methods, each magnet was paired with the two most closely matching schools in the same district (in the sense that their *average* predicted test score or probability of being a magnet in 2004-05 was closest to that of the hypothetical magnet school):

$$(1) \quad Y_{sdt} = \sum_{s=2}^{S} \pi_{0s}\, \gamma_s + \sum_{z=1}^{3} \pi_{1z}\, X_{sdt}^{z} + \sum_{w=2}^{6} \pi_{dw} F_{sdt}^{w} + \pi_7 M_{sdt} + \nu_{sdt}$$

where the π's represent parameters to be estimated, and

Y_{sdt} = the mean outcome for school *s* in district *d* at time *t*.

γ_s = a dummy variable equal to 1 for observations from school *s* and 0, otherwise.

X_{sdt}^{z} = school-level demographic variable z for school *s* in district *d* at time *t.* The three demographic variables are the percentages of students who are: eligible for federally subsidized lunch, African-American, and Hispanic.

F_{sdt}^{w}= dummy variable equal to 1 for observations from district *d* at time *w* if *t* = w, where *w* = 2,…,6. These are simply time dummies for a given district *d*.

M_{sdt} = dummy variable equal to 1 if school *s* in district *d* had magnet status at year *t*, 0 otherwise.

ν_{sdt} = random error term for school *s* in district *d* at year *t*. We allow for correlated error terms across schools within district *d* and over time *t*, through the use of clustering at the district (group) level.

The school-level fixed effects denoted by γ_s allow for stable differences in mean scores between magnets and their comparison schools in the years before and after magnet school conversion. The terms F_{sdt}^{w}*F*are fixed effects that account for variations by year in test scores for magnet/comparison school groups in a given district d. These terms represent year-specific effects common to the magnet and comparison schools within each magnet/comparison group, as shown in Figure 1.

The regressor of interest in the equation is the hypothetical magnet school status indicator variable M_{sdt}. Its coefficient π_7 provides an estimate of whether the "magnet" schools broke from the achievement trajectory of their non-magnet school counterparts in post-interruption years 2004-05 through 2006-07 (i.e., after schools in the treatment group, we pretend, had adopted a magnet school program). This is the coefficient for the CITS treatment effect specified in step 5, above. We also note that the estimated model clusters the errors at the district level to allow for various types of correlation of the error term within district, across schools and especially over time.[16] [17]

Assessing Closeness of Fit

Because the schools we randomly assigned to hypothetical magnet status did not in fact become magnet schools, their post-treatment test-score achievement should, on average, match those of the comparison schools quite closely. Thus, due to random sampling error the treatment group versus -comparison group differences in post-treatment test score changes should appear to be statistically significant in only approximately five percent of the trials, if we reject the null at the five percent level of significance. To the extent that a particular matching method rejects the null hypothesis of equal hypothetical magnet and non-magnet school achievement in more than five percent of the trials, we will conclude that the matching process is not particularly effective.[18]

[16] The study by Bertrand, Duflo & Mullainathan (2002) shows that serial correlation of both the dependent and independent variables in CITS models, where observations are clustered in groups (e.g., districts and points in time), tend to cause underestimated standard errors and subsequently increases the probability of making Type I errors (i.e., rejecting the null hypothesis when it is in fact true). While the outcome model specified above includes both school and year effects, after controlling for these there still may be correlations between individual school-specific observations over years and/or between observations from schools within the same district. The estimation technique accounts for clustering at the district level which, by controlling for the correlation among error terms across years and schools, produces appropriate standard errors.

[17] As mentioned earlier, it is not necessary for the unconditional achievement trajectories for the treatment and comparison groups to be parallel. Rather, we require that the trajectories be parallel after controlling for covariates. However, a small number of CITS studies have used a quite different approach that dispenses altogether with the requirement of parallel trajectories. Instead, these studies parametrically model the time pattern of the outcome variable in the pre-treatment periods, allowing for different patterns between treatment and control schools. These models then extrapolate these divergent patterns into the treatment periods, and interpret divergences from the forecast patterns by the treated school as being due to treatment. For example, Dee and Jacob (2009) model year to year achievement changes by assuming that achievement followed an explicit parametric form (a linear time trend) that differs between the treatment and comparison groups. The validity of this method hinges upon accurate forecasting, using parametric assumptions, of the divergent trends in the treatment period for the treatment and comparison groups. The present study does not use this approach, due to the difficulties inherent in forecasting time patterns into the future. The model used in the present study makes no a priori assumption of what functional form year-to-year achievement changes take, but instead include year-specific indicators (the F_{sdt}^{w} controls in the model presented above).

[18] That is, the method would be prone to Type I errors. Conversely, if a method leads to rejection far less than five percent of the time, that would be cause for concern as well (i.e., it could be prone to Type II errors).

Details of the Analysis

Selection of States, Districts and Potential Schools

To be eligible for inclusion in the study, a state at a minimum had to be consistent in the elementary-level math and ELA tests used over the six-year period from 2001-02 through 2006-07 and provide test scores that could be converted into Z-scores (scores with mean 0 and standard deviation 1 within year and grade based on statewide data). Based on these criteria, we included California, Florida and South Carolina in our study.

In order to test the sensitivity of the results to district size we used the sample of non-magnet elementary schools nationwide in the National Center for Education Statistics (NCES) Common Core of Data, ranked districts by the number of traditional (i.e., non-magnet and non-charter) elementary schools they contained, and chose similarly sized districts in the three eligible states (California, Florida and South Carolina) to include in the analysis. Specifically, from the nationwide sample we eliminated elementary schools that did not operate for the entire six-year period or that were identified as magnet or charter schools based on the CCD. The nationwide set of districts was next ranked according to size as defined by the number of potential (traditional elementary) comparison schools. We then identified the districts in California, Florida, and South Carolina with numbers of traditional schools that most closely matched the 30^{th} through 90^{th} and 95^{th} percentiles of the nationwide distribution of district size. We excluded districts at the 10^{th} and 20^{th} percentiles, which had only one or two schools, because a CITS approach is not feasible with such a small number of schools. To facilitate separate analyses by district size, we used the 55^{th} percentile and the 85^{th} percentile of the national distribution as cutoff points to distinguish between small and medium, and medium and large districts, respectively. Thus, we considered small districts to be those that had four to nine elementary schools that were neither charter nor magnet schools. We labeled as medium districts those with ten to 42 schools, and as large those districts with 43 or more schools that met these criteria.

The resulting sample contained 51 schools from nine small districts, 172 from nine medium districts and 496 from six large districts, for a total of 719 schools across 24 districts in the three states. (Thus, the average number of schools per district was about six, 19 and 83 for small, medium and large districts respectively.)

For a more detailed description of the selection of states, districts and schools for inclusion in the study, please see Appendix F. Appendix F also provides more details on the rationale for selecting 24 districts, and for performing a Monte Carlo analysis on this sample of schools.

Comparing the Relative Effectiveness of the Methods

To gauge the relative success of the various matching methods, we analyzed the estimated CITS treatment coefficients and their corresponding p-values for the 1,000 draws for each estimation method. We report the percentage of times that the null hypothesis of no significant effect was rejected, the mean coefficient on "treated" schools, and the standard deviation of the estimated coefficients, which we can think of as the root mean square error.[19] The first of these statistics is perhaps the most obvious way to assess the alternative approaches because it focuses on statistical significance. Because we have artificially assigned non-magnet schools to hypothetical magnet school status, we would not expect significant differences to arise more than five percent of the time, when we reject the null hypothesis at the five percent level. In turn, rejecting the hypothesis of no effect considerably more than five percent of the time points to poorer performance of the matching method.

We compared the matching methods with a view to identify the method, which rejected the null closest to the expected five percent rate. When this approach failed to identify a single method as best, we used the mean magnet school coefficient and its standard deviation as tiebreakers, in that order. (Better matching methods should produce mean effects closer to zero, and the standard deviation of the magnet coefficient across runs should be smaller.)[20]

Four Variations on a Theme

Variations in the Number of Years of Data

We investigated how the quality of the various matching methods changed as we altered the number of years of test scores included in the pre- and post-treatment periods. This is particularly relevant to analyses of educational achievement because few states have used the same achievement test for more than five or six years at a time. To this end, we repeated the analysis separately for math and ELA scores under the following scenarios:

- three years of both pre-treatment and post-treatment data (3:3)
- two years of pre-treatment and three years of post-treatment data (2:3)
- three years of pre-treatment and two years of post-treatment data (3:2)
- two years of both pre-treatment and post-treatment data (2:2)

Do the Matching Methods Vary in Effectiveness Across District Size?

Second, we varied the size of the districts used. Matching in smaller districts is likely to be relatively difficult because there are fewer schools from which to choose. We hypothesized that the three methods that choose a subsample of potential comparison schools might prove particularly apt in such situations, compared to the method that uses all potential comparison schools. To see this, imagine that there are only five potential comparison schools, none of which may resemble the treated school. But a weighted average of these comparison schools could produce a quite similar match in

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[19] Reported rejections rates are rounded to the nearest tenth of a percent, while both the mean coefficients and standard deviations are reported to the fourth decimal place.
[20] In practice, we found only three ties in rejection rates that called for the use of the second match quality metric (mean coefficient) to determine the best method. All match quality metrics are included in tables in Appendix A.

terms of changes in pre-treatment test scores. For example, if one school had smaller test score growth than the school selected for treatment, and a second had higher test score growth, a weighted average of these two schools could match the treated school's pre-treatment changes reasonably closely. Conversely, a small district might be quite homogeneous demographically and in terms of curriculum and teacher background, so using all traditional schools as the comparison group could in theory work well.

Thus, we repeated the four approaches to matching schools after dividing districts into those we designate small, medium, and large, as defined earlier.

An advantage of showing how well the methods work for districts of varying size is that other researchers contemplating similar analyses can adapt the results to their specific data-set. For example, suppose that a researcher was studying an intervention in elementary schools using a CITS method, and his or her sample consisted of a 2:1 ratio of medium to large districts. By calculating a 2:1 weighted average of the goodness-of-fit statistics, the researcher could reach a conclusion about which approach was the best one for the specific application. In a separate document we will do this explicitly for the NEMS sample of MSAP-funded magnet schools.

Conditioning on Prior Test Scores Only Versus Prior Test Scores and Demographics

In both the regression-based matching and the synthetic control methods, a choice had to be made whether to match schools based solely on lagged test scores, or also on demographic characteristics of the schools. Although lagged scores alone might predict future test scores quite well, additional demographic background variables may also contain useful information with which to predict school-by-school variation in test score changes, even after controlling for past test scores. The appendices contain information on how the regression-based matching and the synthetic control methods were changed to alternatively take into account or exclude demographics of each school's student body (see Appendices B and C). We did not attempt a propensity score model that conditioned only on test scores, because this method is quite distinct in its intent to model the probability of a school becoming a magnet, conditional on all observable data other than outcomes. Put differently, the propensity score method does not seek to focus on the outcome variable, test scores, in the way that the other two methods do.

Matching Separately for Math and Reading Versus Matching Using Average Math and Reading Achievement

It may also prove useful to match based on predicted changes in average math/ELA scores. The advantage of this method is that the set of comparison schools will be identical for the ELA and math analyses. The disadvantage of this approach is that it may produce poor fits for changes in math and/or ELA. To study this issue, we repeated the regression-based matching and the synthetic control methods using the average of the math and ELA Z-scores as the variable upon which matches were made. We also conducted the overall falsification exercise making this average of math and ELA scores the outcome variable of interest, while matching based on lagged values of this average and of demographics.

Results

We summarize the results for each of the four dimensions on which we vary the experiment. The two most important dimensions involve varying the number of years of data used and district size. The other two dimensions, which apply to the regression-based matching and synthetic control methods alone, involve matching based on test scores only versus matching based on test scores and school demographics, and matching based on the average of the Z-scores of math and ELA rather than on the individual subject test scores.

Table 1 reports the percentage of cases in which the CITS model rejected the null hypothesis of equality of the post-treatment changes in test scores for the hypothetical magnet and comparison schools. Each row shows one of the four scenarios of the number of years of pre- and post-interruption data. The rows in the top third of the table show results of the four scenarios for English Language Arts (ELA) test scores. The rows in the middle third show results for math scores. The bottom panel of rows show results when we model the average of math and ELA scores, and match using pre-interruption averages of ELA and math scores, as well as student demographics. The columns from left to right show "All Districts" results in which we use all 24 districts for each of the four approaches to identifying comparison schools, followed by the results when our simulations for each approach are run separately for nine small, nine medium and six large districts.

Results When Combining All 24 Districts

Beginning with models of ELA achievement scores using all 24 districts, shown in the upper left of Table 1, we find that all four methods over-reject the null hypothesis of no magnet effect, but typically by small amounts. That is, at the five percent level of statistical significance we expect to reject the null hypothesis of no magnet effect in five percent of cases. This is because none of our hypothetical treatments actually occurred. Thus, an approach that rejects the null more than five percent of the time is "over-rejecting". To aid the reader in ranking the four methods, we include symbols, ranging from white circles showing the matching method that rejects closest to five percent of the time to completely black circles showing the method that diverges the most from the five percent level.[21] Ties are indicated when two or more methods share the same symbol.

The method that uses all traditional schools in the district as a comparison, thereby avoiding a deliberate matching of treated and comparison schools, performs the best for examining changes in ELA achievement scores in the sense of rejecting most closely to five percent of the time. Results vary by the number of available years of pre- and post-interruption data, with the rejection rate of the null hypothesis for the "all traditional schools" comparison method ranging from 6.3 to 7.6 percent of the

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[21] A circle that is three-quarters white indicates the method that is second closest to rejecting the null five percent of the time. A circle that is half-black and half-white indicates the method ranks third among the four approaches.

1,000 replications. The method that is second closest to rejecting at the nominal five percent rate is the regression-based matching approach, which over-rejects just slightly more than the "all comparison schools" approach. The synthetic control and propensity score models perform similarly to each other, and depending on the number of years of pre-interruption and post-interruption data we use, these two methods reject the null roughly 7 to 9 percent of the time. Close inspection of the data suggests that the propensity score model over-rejects slightly more than the synthetic control model.

The results for models of math scores, using all districts, shown in the middle rows, and the leftmost quarter of the table, are roughly similar. For the method that uses all traditional schools as the comparison, rejection rates range from 5.6 to 7 percent for different scenarios of available data. Regardless of how many years of "pre" and "post-interruption" data are included, the method that uses all traditional schools as the comparison group performs the best in the sense of over-rejecting the least. The propensity-score models perform second best by this metric, followed by the regression-based matching method and finally the synthetic control method. The synthetic control method over-rejects particularly often in the case of math. One way of seeing this is to examine the maximum rejection rates across the four combinations of pre- and post-interruption years, which for the "all traditional schools", propensity, regression-based and synthetic methods are 7.0, 7.6, 8.7 and 11.6 percent, respectively.

Turning to models of the *average* of math and ELA scores, in the bottom panel, the "all traditional schools" and regression-based matching methods over-reject by about the same amount. These are the top-ranking methods by the criterion of the least amount of over-rejection, followed by the propensity-score models, and finally, the synthetic control method.[22]

As a general indication of how each method performed across all of the scenarios defined by combinations of subject and pre/post data availability, the final row of Table 1 contains the average rejection rate for each column of the table. For example, the first column of the table shows that using the "All Districts" sample with the all comparison schools method resulted in an average rejection rate across all scenarios of 6.7 percent, while the synthetic control group method using the same sample yielded an 8.6 percent rate.

Results for Models Estimated Separately by District Size

To the right of the "all districts" models in Table 1, we show the results we obtained when we modeled test scores for the subsamples of small, medium and large districts. We will spare the reader a detailed rundown on the results, and instead will focus on broad patterns.

Perhaps the most striking aspect of these panels in Table 1 is how high the rejection rates for the "Small", "Medium", and "Large" districts samples are compared to the full "All Districts" sample. This discrepancy most likely arises due to the small number of districts in the three subsamples. Imbens and Wooldridge (2007) show that clustering of standard errors does an effective job of controlling for

[22] For the case with three years of pre- and post-treatment data, there was a tie between the "use all" and regression-based matching methods.

within-group correlation of error terms when the number of groups is quite high. However, they show that when the number of groups is small, the standard errors tend to be biased downward, leading to an over-rejection of the null hypothesis. Bertrand, Duflo, and Mullainathan (2002) make a similar point using a simulation, and they emphasize that serial correlation can be an important source of this within-group correlation. In the small, medium and large district subsamples, we have only 9, 9, and 6 districts respectively. In this case we have probably failed to fully take into account correlations across observations within districts.

Setting aside the high absolute percentage rate of rejection in each of these smaller samples, how do the four methods of selecting comparison schools perform relatively to each other? The results are mixed across types of test scores modeled for small districts, but quite consistent and clear-cut for the medium and large districts.

In <u>small districts</u>, the regression-based matching approach appears to work best for ELA test scores in the sense of rejecting the closest to five percent of the time, while the propensity-score model comes closest to rejecting five percent of the time for models of math scores and the average of math and ELA scores. The "all traditional schools" method performs almost as well as the propensity-score and regression-based matching schemes, whereas the synthetic control method typically over-rejects the null hypothesis for all three outcomes (i.e., ELA, Math and average ELA/Math) to a greater degree than the three other methods.

In <u>medium-sized</u> districts, regardless of which measure of achievement we model, the "all-traditional schools" approach performs best, followed closely by the regression-based matching method. The synthetic-control and propensity-score models over-reject the most often, and the gap in performance with the two better performing methods is large, with these lower-performing methods over-rejecting by typically two to five percentage points more than the other two methods.

In <u>large districts</u>, using all traditional schools as the comparison group again leads to the least degree of over-rejecting the null hypothesis. The synthetic control approach comes in second, and the regression-based and propensity-score methods roughly tie for last place. The gap between the "all traditional school" approach and the three other approaches in the percentage of cases in which the null hypothesis of no magnet effect is rejected is always at least one percentage point and typically two or more percentage points.

Overall, with some exceptions, the analyses for subsamples of districts suggests that the "all traditional school" comparison group is the best approach in that it over-rejects the null hypothesis the least, with the exception of small districts, for which the propensity score approach or in the case of ELA achievement the regression-based matching method most often rejects closest to the expected five percent rate. The synthetic control method typically over-rejects more than any of the other methods. The case in which the synthetic control method appears to come in second place to the all traditional schools approach and outperform both regression-based matching and propensity score methods is for large districts, where we have the smallest sample of district clusters, at six, and the largest number of schools per cluster.

For researchers using a different mix of district sizes, it is likely that the "all traditional school" approach will over-reject the least of the four methods, with a possible exception being a sample with a heavy concentration of small districts. It is likely that any of the other three methods could be the second-best method in terms of over-rejecting the least, depending on the distribution of district sizes in the researchers' actual sample.

Other Metrics: Mean Coefficient and Root Mean Square Error

We have focused on the percentage rejection rate as the primary metric by which we would rank methods. It is useful, though, to examine how the various methods perform if we instead rank methods either by the average coefficient on the magnet school dummy or the standard deviation of this coefficient. Appendix Table A-1A shows all three criteria side by side (the magnet coefficient rejection rate, the mean of the magnet school coefficients, and the root mean square error of the magnet school coefficients), for the "All Districts" sample, for each scenario of years of data available, and each matching method. This way of organizing the table in some sense compares apples to oranges, as these are different metrics, but our foremost goal in this table is to check whether our secondary methods of comparing models tell approximately the same story. For this purpose, the circle symbols that indicate the ranking of methods also prove helpful.

Examination of the absolute value of the mean coefficient (we take an average of the magnet school coefficients, and then the absolute value of this average) and the standard deviation of the coefficients (root mean square error) for the sample of all districts supports the conclusions we reached above. We first consider the mean of magnet coefficients, where the regression-based matching was shown to rank as the best method in 5 of the 12 scenarios, but it also ranked third best in 5 scenarios and worst possible method in 1 scenario. In comparison, while the all traditional public schools only ranked as the best method three times, in half of the scenarios it was considered second best. In this case it is useful to compare the average value of the quality metric across all tested scenarios for each method, which shows the all traditional public schools to perform best with the lowest average mean of 0.0031. Choosing a second best approach according to the mean magnet coefficient is not as easy, however. Here, the results show that the regression-based and propensity score methods worked equally well (each had an average of mean magnet coefficients across scenarios equal to 0.0038).

Using the root mean square error criterion, the method of using all traditional public schools was deemed best in 10 of the 12 scenarios defined by test subject and years of pre/post data availability. The average of the calculated root mean square errors across scenarios was the smallest for this method as well (equal to 0.0239). The second-best approach according to this quality metric was the synthetic control group method, which was ranked most effective in 2 of the 12 scenarios and took second place in the remaining 10 scenarios.

Appendix Tables A-1b, A-1c, and A-1d show the similar three metrics across the scenarios across the subject/data availability scenarios for the samples of small, medium and large districts. Rather than describe each of these tables in detail Appendix Table A-1e provides a summary of the average metrics across the scenarios included in the last row of each of these tables. In addition, the last row of

Appendix Table A-1e presents the grand mean of the average metric figures calculated in the previous tables. Based upon the grand means it is clear that the preferred method according to all three quality metrics is the method that uses all traditional public schools. However, identifying a second-best method is not as straightforward; all three quality metrics provide a different answer. According to the magnet coefficient rejection rate, the second most effective method would be regression-based matching. The mean of the magnet coefficients suggests that the second-best method is propensity score matching. Finally, using the root mean square error of the magnet coefficients the synthetic control group method is deemed second best.

Variants of the Regression-Based Matching and Synthetic Control Methods

Conditioning on Test Scores Alone

In addition to variations by district size and years of data used, another dimension along which the methods might differ is the covariates used in matching. For both the regression-based matching and synthetic control methods, matching using only the lagged dependent variable (test scores) is clearly an option. Does this approach fare any worse than in the above results, for which we conditioned not only on pre-interruption test scores but also demographics? (As mentioned earlier, it makes little sense to re-do the propensity score models conditioning on test scores alone, because racial and socioeconomic isolation is clearly a factor that has influenced the decision to create magnet schools in the real world.) Although researchers have relatively little experience with the synthetic control method due to its recent introduction, the voluminous literature on regression-based forecasting provides strong hints that adding more regressors to a forecast can lead to "overfitting". For instance, macroeconomic forecasts based on Unrestricted Vector Autoregression (UVARS), in which each variable in a system is regressed on several lags of its own values and the values of the other variables, leads to poor and erratic forecasts that are particularly sensitive to changes in one or two variables in the system. (See for example the review by Todd (1990).) In our context, this would translate as precision of estimates potentially falling, and the confidence intervals of forecasts becoming larger, when we add the demographic variables to the regression-based forecasts.

Table 2a repeats the results from the "All Districts" analyses in Table 1, but additionally shows corresponding data when we repeated the analyses after using only pre-interruption test scores to perform the matching. Our random selections of magnet schools in the 1,000 replications are identical in the two panels – all we change is the number of variables used in the matching processes. To facilitate comparisons, for each pair of variations on a given method, we shade in gray the variation that performs better in terms of over-rejection.

Table 2a reveals that the longstanding criticism of UVAR models may apply in our context: the over-rejection rate in the regression-based matching models is smaller in the models that control only for lagged test scores than in the models that control for lagged test scores plus three demographic variables in ten of 12 cases. The rightmost panel of Table 2a shows corresponding results for the synthetic control approach. The results here are quite different, perhaps because the synthetic control method is not a regression-based method – the method that includes demographics leads to lower over-rejection rates than the method that excludes demographic variables in every case. If we had to choose among the four methods displayed in Table 2a, in nine of 12 cases the best performer, in the sense of rejecting the closest to five percent of the time, is regression-based matching that conditions on lagged test scores but not on demographic variables.

Table 2b shows the corresponding rejection rates of both methods for the small, medium and large districts when using test scores and demographics versus test scores alone. No strong patterns emerge in this table. The advantages we noted in Table 2a of dropping the demographic variables from the regression-based matching method appear to stem mostly from the large districts, in which case the regression-based method that matches based on lagged scores alone outperforms the version that includes demographics as well as test scores in two-thirds of cases, and sometimes markedly so.

Appendix Tables A-2a for the regression-based approach and A-3a for the synthetic control method provide expanded versions of Table 2a that include all three metrics by which we can compare the matching methods.[23] Appendix Tables A-2b, A-2c, and A-2d for the regression-based approach and Appendix Tables A-3b, A-3c, and A- 3d for the synthetic control method show all three metrics for small, medium and large districts. These tables have been included mainly so that other researchers can take a weighted average of results across district size to fit their own particular samples.

Conditioning on Average of ELA and Math Scores

Our second variation experiments with the achievement criteria upon which we match. It is quite common in models of education interventions to measure the impact on both math and ELA scores. Both the "all traditional schools" method and the propensity score method pick identical comparison groups for models of math and ELA scores. In contrast, the regression-based and synthetic control group methods will produce different matches for a given magnet depending on which achievement measure we are modeling. Imagine that an evaluation found that magnet schools boost math achievement but have no effect on ELA scores. One possibility might be that the difference in the findings stems not from true differences in effects, but rather from differences in the comparison groups compiled for the math and ELA analyses. If a researcher had strong reservations along these lines, he or she might prefer to choose an identical comparison group for both math and ELA analyses.

But this alternative approach is potentially misleading, because it is likely that the comparison schools selected in this way could be worse matches than the matches performed specifically for math or for ELA scores individually. If the matches are poor, we are likely to over-reject the null of no magnet school effect because the key condition for the CITS models to produce unbiased estimates – similar

[23] The two regression-based methods perform similarly (and better) than the synthetic control method when we consider the mean coefficient. For instance, in nine cases out of 12 one of the two regression methods produces the lowest mean coefficient, and in 10 of 12 cases one of these methods produces the second-lowest coefficient. But the root mean squared errors are the lowest for the two approaches using the synthetic control method. The two variants of the synthetic control method always rank first or second by this method. Notably, the regression-based method that conditions on lagged test scores and demographic variables produced the largest root mean square error in all cases but one out of twelve, which probably reflects the imprecision caused by overfitting.

year to year changes in the outcome of interest between the magnet and the comparison schools – is likely to be violated.

We examined this approach, and the possible over-rejections that it would induce, by repeating the regression-based matching and synthetic control models in Table 1, but matching using the average of math and ELA scores in each year. We then use the single set of matches generated to model the effect of magnet schools on math and ELA achievement separately.

Table 3 reports the results using the "All Districts" sample. The column labeled "Regression-Based Matching" on the left repeats the results from Table 1, and provides a comparison for the next column, which shows the results when we matched using the average of math and ELA scores. In the case of ELA scores, we find that for all combinations of years, the percentage rejection rate is closer to five percent for the model that matches based on the average of math and ELA scores. When we model math achievement, the model that matches based on the average of the ELA and math scores over-rejects less in two of four cases, and by less on average. It is possible that in spite of our concerns that schools could have disparate changes in ELA and math, the additional precision afforded by averaging over the two test scores may outweigh this disadvantage.

The rightmost pair of columns in the table compare the synthetic control group results from Table 1 with the results when matching are based on the average of ELA and math scores. No clear pattern emerges. The latter method has a lower rejection rate in four cases out of eight.

Appendix Tables A-4a for the regression-based approach and A-5a for the synthetic control method show versions of Table 3 that include the other two metrics for selecting a method (magnet coefficient and root mean square error). Appendix Table A-4a reveals no clear winner when we compare the two versions of regression-based matching in terms of mean coefficient and root mean square error. For regression-based matching, either method has the "better" outcome in terms of mean coefficient or root mean square error in about half of cases. Appendix Table A-5a shows that the version of the synthetic control method that averaged prior ELA and math scores resulted in a lower mean coefficient for all four scenarios in math, but for only one of the four in ELA. The version of the synthetic control group approach that conditioned on the average of prior ELA and math scores performed worse with respect to root mean square error than the version that conditioned on prior tests in the same subject area as the achievement outcome being modeled. In all four scenarios in both ELA and math the root mean square error was larger for the version of the synthetic control group approach that conditioned on the average of prior ELA and math scores.

Appendix Tables A-4b, A-4c, and A-4d for the regression-based approach and Appendix Tables A-5b, A-5c, and A-5d for the synthetic control method repeat these analyses for small, medium and large school districts and show all three metrics for each model. The tables are particularly intended for the use of researchers who might be interested in choosing among the methods, but using a different mix of district sizes. The results are complex. Appendix G provides discussion of the results for districts of different sizes for the interested reader.

Robustness to Clustering

Clustering the standard errors at the district level appears to be important in yielding results that approach the expected five percent rejection rate. We repeated the analyses in Table 1 for the "all-district" sample, without clustering. Table 5 shows the results. Consistent with Bertrand, Duflo, and Mullainathan (2002), we over-reject the null hypothesis of a zero magnet school effect considerably more often without clustering. When we cluster, as in Table 1, rejection rates vary from 5.6 to 11.6 percent. When we fail to cluster, as in Table 6, rejection rates range from 7.5 to 18.8 percent. The average rejection rate for the "All Districts" sample without clustering in Table 5 was 12.8 percent. This figure is well above the 7.6 percent rejection rate when averaged across the corresponding results in Table 1, in which clustering at the district level was used.

Conclusion

This paper yields information on the relative ability of four different methods of estimating the effect of interventions on schools using a comparative interrupted time series (CITS) approach, also known as a difference-in-differences approach. The first method, in which all traditional (i.e., untreated) schools are used as the comparison group, is an approach often used by economists; it explicitly avoids attempts to match treatment and comparison schools at all. The second method, which is regression-based, uses regression models to predict which schools among the potential control group are likely to have post-interruption achievement changes similar to those that the treated schools would experience in the absence of treatment. The third method uses propensity score modeling of the probability that schools become MSAP-funded magnet schools to match schools. The final method, the synthetic control group method, is somewhat similar to the regression-based approach, but takes a weighted average of two or more schools and treats them as a single composite comparison school.[24]

Overall, for the mix of district sizes that we selected, it appears that the method that came the closest to the expected five percent rejection rate was the first method, which uses all traditional schools as the comparison group. We note that other researchers might come to different conclusions based on their mix of district sizes. Notably, in small districts the preferred methods were the propensity score approach for modeling either math test scores or an average of ELA and math scores and the regression-based matching in the case of ELA test scores. In these cases, the all traditional schools comparison method rejected the null hypothesis more often than these preferred methods. In contrast to district size, which appears to matter considerably, we did not typically find large systematic variations in the ranking of methods across the number of years of pre- and post-treatment data available.

Why did the all traditional schools approach that uses all potential comparison schools generally work better than the three matching methods? One part of the answer may be that adding more

[24] Our approach was not to take a weighted average but instead to run a weighted least squares model with weights of potential comparison schools proportional to the weights produced by the synthetic control method.

comparison schools increases the precision of the estimates. Another important part of the answer may be that the various matching methods do not in all cases reduce bias to the extent one might expect.

Consider first the idea of taking an average across all potential comparison schools, which is in effect how the "all traditional schools" method works. The key requirement for the comparative interrupted time series model to produce an unbiased estimate is that in the treatment period the counterfactual pattern of test scores for the treated school, had it not been treated, should be parallel to the pattern of test scores observed for the comparison schools. Each of the three methods that explicitly attempts to match does so based on the values of variables in the pre-treatment period only. They will produce biased results if the comparison schools selected based on the pre-treatment data do not track the counterfactual post-treatment pattern of test scores for the treated school. Matching based on pre-treatment values of variables does not guarantee a good match in the post-treatment period.

The all traditional schools method that uses all potential comparison schools may reduce the chances of bad matches by in effect averaging outcomes across many comparison schools. This approach may succeed simply because unusual fluctuations in year-to-year changes in the outcome variable among comparison schools in the treatment period tend to average out as the number of comparison schools rises. Put differently the variance of the average post-treatment patterns in the outcome variable for the comparison group will fall as the number of comparison schools rises. This idea is supported by the fact that this method produced the smallest root mean square error in most cases.

Second, consider the second half of the explanation: the three methods that explicitly attempt to match may not work very well, and indeed, they can be particularly prone to omitted variable bias in the treatment period when the number of selected comparison schools is quite small. In the case of regression-based matching and propensity score matching, the number of comparison schools is indeed small.

In the case of regression-based matching, for instance, we saw that the models that conditioned on past values of the outcome variable tended to over-reject less than the corresponding models that additionally conditioned on lagged demographic variables. The higher rejection rate of the model that also includes lagged demographic variables is indicative of over-fitting. The idea here is that collinearity between the regressors can produce unreliable and large coefficients, such that values of the predicted future outcome variable can become rather sensitive to small changes in the values of the regressors.

The results also showed that the propensity score model did not fit very well. This suggests a high degree of imprecision in matching based on the probability that a school adopted a magnet program, which increased the chances of a poor match.

The synthetic control method did not in general outperform either the regression-based or propensity score matching approaches except in the case of large districts, where it performed almost as well as comparison approach that used all traditional schools. One reason for the increased relative performance of the synthetic control method in large districts (in which the mean number of schools was 83) is that larger districts are almost by definition more heterogeneous along multiple dimensions than small neighborhood districts. Finding good matches along multiple dimensions, including achievement and demographics, becomes more important in the context of larger districts (and better matches within all that heterogeneity are more easily found).

Although the all traditional schools method that uses all potential comparisons has in general performed the best in the sense of over-rejecting the null hypothesis the least often, we note that in all cases the rejection rate exceeded five percent. This over-rejection almost surely reflects the finding by others that CITS models with only a few treatment and comparison groups tend to reject the null hypothesis of a zero effect too often, because serial correlation within groups is mistaken for a true causal effect of a reform that occurs at roughly the same time as a large random shock. For this reason, even if researchers use the "all traditional schools" approach, it may not be enough to cluster the error terms at the group (district) level to solve this problem. This issue is minor for our "All Districts" sample of 24 magnets in 24 districts. For instance, depending on the number of pre- and post-interruption years, our rejection rates were 6.3 to 7.6 percent for ELA models and 5.9 to 7 percent for math models using all comparison schools with the all-district sample.

But for researchers using a smaller number of districts, clustering by itself will not be sufficient. This can be seen by examining the rejection rates for models run on the samples of nine small districts, nine medium districts, and six large districts separately, where even our preferred method rejects far above the expected 5 percent rate (on the order of 8.2 to 14.5 percent). In such cases additional approaches, such as using bootstrapping of the evaluation sample may be necessary. See Cameron, Gelbach and Miller (2008), for a recent review and extension to these approaches.

These results could also prove informative for a wide variety of CITS settings where subsamples of comparison schools are being chosen. The authors of this paper plan to post a downloadable spreadsheet that will allow other researchers to analyze, for their particular mix of district sizes and years of data availability, the relative rankings of the four main methods, and the several variants that we have examined in this paper.

For researchers working in the context of evaluating school reforms, the main lesson from these simulations is that we may need to reconsider the quite common approach of selecting subsamples of comparison schools in CITS models. Indeed, researchers may do more harm than good when estimating CITS models by selecting a subsample of the untreated schools within the district as the comparison group. Quite often, using all potential comparison schools in the given district may prove more reliable. The extent to which this is true depends, however, on the mix of the sizes of the districts in terms of the number of potential comparison schools.

Table 1 – Percentage of Trials in Which the Null Hypothesis of No Difference in the Three-Year Average Post-Interruption Change in Test Scores between Magnet and Comparison Schools is Rejected at the 5% Level of Significance by Combination of Pre/Post Test Data, District Size and Matching Approach

Subject	Years of Pre/Post Data	ALL DISTRICTS				SMALL DISTRICTS				MEDIUM DISTRICTS				LARGE DISTRICTS			
		All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group
ELA	3-3	○ 6.5	◑ 7.3	● 7.8	◔ 7.2	● 11.3	◔ 9.2	○ 8.9	◑ 10.2	◔ 9.8	○ 9.6	● 14	● 14	○ 13.6	◑ 18.4	● 18.5	◔ 15.1
	3-2	○ 6.3	◔ 7.5	◑ 7.7	● 8.2	○ 9.5	◔ 9.6	◑ 9.8	● 11.4	◔ 9.5	○ 9.3	◑ 12.6	● 14.4	○ 14	● 18.3	◑ 16.1	◔ 15.4
	2-3	○ 7.6	◑ 8.2	● 9	◔ 7.7	◑ 13.1	◔ 13	● 14	○ 12.3	○ 8.2	◔ 11.4	◑ 12.8	● 14.9	○ 14.3	● 19.1	◑ 16.7	◔ 16.3
	2-2	○ 6.5	◔ 7.9	● 8.7	◑ 8.5	○ 9.9	◔ 11.1	● 12.5	◑ 12.1	○ 10.1	◔ 12.8	◑ 13.4	● 14.1	○ 14.5	● 19	◑ 15.9	◔ 15.7
MATH	3-3	○ 6.9	◔ 7.2	◑ 7.6	● 11.6	◑ 11.1	◔ 10.1	◔ 10.1	● 12.8	○ 9.9	◔ 12.5	◑ 13.9	● 18.6	○ 12.6	◑ 17.5	● 18.1	◔ 15.5
	3-2	○ 5.9	◑ 7.5	◔ 6.1	● 9.3	◑ 10.5	◔ 10.3	○ 9.3	● 11.6	○ 10.7	◔ 11.9	● 15.5	◑ 15	○ 13.9	◔ 16.4	● 18.3	◑ 16.6
	2-3	○ 7	◑ 8.7	◔ 7.6	● 8.9	◑ 12.2	● 16.7	○ 10.3	◔ 11.5	○ 10.3	◔ 11.7	◑ 14	● 16.4	○ 13.1	● 17.5	◑ 17.2	◔ 15.2
	2-2	○ 5.6	◑ 7.3	◔ 6.1	● 7.5	◑ 11.7	◔ 11.4	○ 10.3	● 14.3	○ 9.6	◔ 11.8	● 14.1	◑ 13.1	○ 13.5	◑ 18	● 18.3	◔ 15.6
Average ELA/MATH	3-3	◔ 6.9	◔ 6.9	◑ 7.5	● 9.3	◑ 10.3	◔ 9	○ 8.7	● 13.3	○ 10	◔ 10.9	◑ 15.8	● 15.9	○ 13	● 18	● 18	◔ 14.1
	3-2	◔ 7	○ 6.5	◑ 7.9	● 9.3	◑ 9.5	○ 8.1	◔ 8.3	● 14.4	○ 10.8	◔ 12.7	● 14.3	◑ 13.5	○ 12.9	● 17.4	◔ 16.6	◑ 17.3
	2-3	◔ 7.5	● 8.3	◑ 8.1	◔ 7.5	◔ 12	● 16.3	○ 11.8	◑ 13.1	○ 9.2	◔ 11.8	● 14.4	◑ 14	○ 14.1	◑ 16.3	● 16.5	◔ 15.2
	2-2	○ 6.9	◔ 7.6	● 7.9	◑ 7.7	○ 10.8	● 16	◔ 11.5	◑ 12.3	○ 9.4	◔ 11.5	● 13.4	◑ 13	○ 11.8	● 18	◑ 17.8	◔ 15.7
Average Across Subject and Years of Pre/Post Data		○ 6.7	◔ 7.6	◑ 7.7	● 8.6	◔ 11.0	◑ 11.7	○ 10.5	● 12.4	○ 9.8	◔ 11.5	◑ 14.0	● 14.7	○ 13.4	● 17.8	◑ 17.3	◔ 15.6

Note: In each row and subset of district size, the circles indicate the ranking of each method. The white circle indicates the best method, in the sense that the percentage of runs in which the null was rejected came the closest to 5%, and the black circle indicates the worst method. A circle that is three-quarters white indicates the method that is second closest to rejecting the null five percent of the time. A circle that is half-black and half-white indicates the method ranks third among the four approaches.

Table 2a – Regression-Based and Synthetic Control Models Using All Districts With and Without Demographics as Covariates

Subject	Years of Pre/Post Data	ALL DISTRICTS			
		Regression-Based Matching With Demographics as Covariates	Regression-Based Matching Without Demographics as Covariates	Synthetic Control Group With Demographics as Covariates	Synthetic Control Group Without Demographics as Covariates
ELA	3-3	7.3	7.5	7.2	8.8
	3-2	7.5	6.3	8.2	9.5
	2-3	8.2	7.3	7.7	8.0
	2-2	7.9	9.0	8.5	8.7
MATH	3-3	7.2	6.7	11.6	12.4
	3-2	7.5	7.0	9.3	11.7
	2-3	8.7	6.8	8.9	9.6
	2-2	7.3	6.0	7.5	9.8
Average ELA/MATH	3-3	6.9	6.5	9.3	9.6
	3-2	6.5	6.3	9.3	10.2
	2-3	8.3	7.6	7.5	8.3
	2-2	7.6	6.9	7.7	8.5
Average Across Subject and Years of Pre/Post Data		7.6	7.0	8.6	9.6

Note: Table shows the percentage of trials in which the null hypothesis of no difference in the two or three year average post interruption change in ELA, Math, and average ELA/Math scores between magnet and comparison schools is rejected at the 5% level of significance when matching is based on pre-interruption test scores and student demographic characteristics vs. only pre-interruption test scores. The results are shown for different combinations of years of per- and post-interruption data and by the regression-based and synthetic control matching approaches. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject and years of pre/post data.

Table 2b – Regression-Based and Synthetic Control Models Using All Districts With and Without Demographics as Covariates, by District Size

Subject	Years of Pre/Post Data	SMALL DISTRICTS				MEDIUM DISTRICTS				LARGE DISTRICTS			
		Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores
ELA	3-3	9.2	9.5	10.2	11.4	9.6	11.9	14.0	15.3	18.4	18.5	15.1	15.2
	3-2	9.6	9.7	11.4	13.7	9.3	11.5	14.4	15.4	18.3	18.9	15.4	16.8
	2-3	13.0	12.6	12.3	12.4	11.4	12.4	14.9	13.8	19.1	18.4	16.3	17.2
	2-2	11.1	11.5	12.1	11.8	12.8	14.8	14.1	14.1	19.0	19.3	15.7	18.3
MATH	3-3	10.1	11.0	12.8	10.0	12.5	11.6	18.6	14.5	17.5	17.1	15.5	17.0
	3-2	10.3	13.1	11.6	12.6	11.9	11.6	15.0	12.4	16.4	15.9	16.6	17.3
	2-3	16.7	13.8	11.5	11.2	11.7	11.8	16.4	14.8	17.5	16.0	15.2	17.1
	2-2	11.4	11.9	14.3	12.4	11.8	11.2	13.1	11.8	18.0	15.9	15.6	15.8
Average ELA/MATH	3-3	9.0	8.7	13.3	10.5	10.9	10.8	15.9	13.8	18.0	14.7	14.1	14.6
	3-2	8.1	7.9	14.4	12.2	12.7	11.0	13.5	13.4	17.4	16.5	17.3	16.4
	2-3	16.3	13.5	13.1	12.2	11.8	11.4	14.0	13.4	16.3	16.6	15.2	15.9
	2-2	16.0	13.0	12.3	10.7	11.5	11.6	13.0	12.1	18.0	17.1	15.7	16.2
Average Across Subject and Years of Pre/Post Data		11.7	11.4	12.4	11.8	11.5	11.8	14.7	13.7	17.8	17.1	15.6	16.5

Note: Table shows the percentage of trials in which the null hypothesis of no difference in the two or three year average post interruption change in ELA, Math, and average ELA/Math scores between magnet and comparison schools is rejected at the 5% level of significance when matching is based on pre-interruption test scores and student demographic characteristics vs. only pre-interruption test scores. The results are shown for small, medium, and large districts by different combinations of years of pre- and post-interruption data and by the regression-based and synthetic control matching approaches. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject and years of pre/post data. Ties are signified by neither element of the pair being highlighted.

Table 3 – Regression-Based Matching and Synthetic Control Models for ELA and Math for "All-District" Sample (from Table 2) and Repeated Using Matches Based on the Average of ELA and Math Scores

Subject	Years of Pre/Post Data	ALL DISTRICTS			
		Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group - Using Average of ELA and Math
ELA	3-3	7.3	6.3	7.2	8.0
	3-2	7.5	6.9	8.2	9.4
	2-3	8.2	7.3	7.7	6.9
	2-2	7.9	6.7	8.5	9.0
MATH	3-3	7.2	6.8	11.6	10.1
	3-2	7.5	6.0	9.3	9.5
	2-3	8.7	8.8	8.9	8.1
	2-2	7.3	7.8	7.5	7.4
Average Across Subject and Years of Pre/Post Data		7.7	7.1	8.6	8.6

Note: Numbers in cells indicate percentage of trials in which the null hypothesis of no difference in the two or three year average post-interruption change in ELA and math scores between magnet and comparison schools is rejected at the 5% level of significance by four scenarios of pre/post test data availability and matching approach (regression-based approach using ELA/math separately versus using average of ELA and math). Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject and years of pre/post data.

Table 4 – Version of Table 1 Results for "All-District" Sample without Clustering at the District Level

Subject	Years of Pre/Post Data	ALL DISTRICTS			
		All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group
ELA	3-3	○ 17.5	◑ 17.6	● 18.8	◑ 17.6
	3-2	◑ 12.0	◔ 11.5	● 12.6	◔ 11.5
	2-3	● 18.0	◑ 16.8	○ 16.3	◑ 16.8
	2-2	● 12.4	◔ 9.8	◑ 11.2	◔ 9.8
MATH	3-3	● 16.3	◔ 14.8	◑ 16.0	◔ 14.8
	3-2	● 10.3	◔ 9.0	◑ 10.2	◔ 9.0
	2-3	○ 10.5	◑ 12.3	● 14.4	◑ 12.3
	2-2	○ 7.3	◑ 7.6	● 9.1	◑ 7.6
Average ELA/MATH	3-3	◑ 16.2	◔ 15.8	● 18.4	◔ 15.8
	3-2	◑ 11.1	◔ 9.8	● 12.4	◔ 9.8
	2-3	○ 12.5	◑ 13.5	● 15.7	◑ 13.5
	2-2	○ 7.5	◑ 8.8	● 10.9	◑ 8.8
Average Across Subject and Years of Pre/Post Data		12.6	12.3	13.8	12.3

Note: In each row and subset of district size, the circles indicate the ranking of each method. The white circle indicates the best method, in the sense that the percentage of runs in which the null was rejected came the closest to 5%, and the black circle indicates the worst method. A circle that is three-quarters white indicates the method that is second closest to rejecting the null five percent of the time. A circle that is half-black and half-white indicates the method ranks third among the four approaches.

References

Abadie, Alberto and Javiet Gardeazabal, (2003), "The Economic Costs of Conflict: A Case Study of the Basque Country", *American Economic Review*, vol. 93, no. 1, 112-132.

Abadie, Alberto, Alexis Diamond and Jens Hainmueller, (2007), "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program," National Bureau of Economic Research, Working Paper 12831, January.

Betts, Julian, Jesse Levin, Marian Eaton and Bruce Christenson, (2008), "Analysis Plan for National Evaluation of Magnet Schools," American Institutes for Research, prepared for Institute of Education Sciences, National Center for Education Evaluation and Regional Assistance , U.S. Department of Education, Contract Number: ED-04-CO-0025/009, October.

Betts, Julian R., Lorien A. Rice, Andrew C. Zau, Y. Emily Tang, and Cory R. Koedel, 2006), *Does School Choice Work? Effects on Student Integration and Achievement*, San Francisco: Public Policy Institute of California. Available from www.ppic.org.

Bloom, Howard, Fred Doolittle, Michael Garet, Bruce Christenson, and Marian Eaton (2004), *Designing a Study of the Impact of Magnet Schools on Student Achievement: Alternative Designs and Tradeoffs,* MDRC and AIR, submitted to the U.S. Department of Education Institute of Education Sciences Under Task Order #ED-01-CO-0060/0002, July.

Bloom, Howard S., JoAnn Rock, Sandra Ham, Laura Melton and Julieanne O'Brien (2000), Evaluating the Accelerated Schools Approach: A Look at Early Implementation and Impacts on Student Achievement in Eight Elementary Schools. New York. MDRC.

Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller (2008), "Bootstrap-Based Improvements for Inference with Clustered Errors," *Review of Economics and Statistics*, (90:3), August, pp. 414-427.

Card, David (1990), "The Impact of the Mariel Boatlift on the Miami Labor Market", *Industrial and Labor Relations Review*, 43:245-257.

Bertrand, M., Duflo, E. and Mullainathan, S. (2002). "How Much Should We Trust Differences-in-Differences Estimates?" NBER Working Paper No. 8841.

Dee, Thomas and Brian Jacob (2009), "The Impact of No Child Left Behind on Student Achievement", NBER Working Paper No. 15531.

Imbens, Guido and Jeffrey Wooldridge (2007), "*What's New in Econometrics? Cluster and Stratified Sampling",* Lecture Notes 8, National Bureau of Economic Research, Summer 2007.

Neumark, David and William Wascher (1992), "Employment Effects of Minimum and Subminimum Wages: Panel Data on State Minimum Wage Laws", Industrial and Labor Relations Review, (46:1), October.

Shadish, William R., Thomas D. Cook, Donald T. Campbell (2002), Experimental and Quasi-Experimental Designs for Generalized Causal Inference, Boston: Houghton Mifflin.

Todd, Richard M. (1990), “Improving Economic Forecasting with Bayesian Vector Autoregression,” in C. W.J. Granger (Ed.) *Modelling Economic Series*, New York: Oxford University Press.

Wolf, Patrick, Babette Gutmann, Michael Puma, Brian Kisida, Lou Rizzo, and Nada Eissa (2009, *Evaluation of the DC Opportunity Scholarship Program: Impacts After Three Years* (NCEE 2009-4050). Washington, DC: National Center for Education Evaluation and Regional Assistance, Institute of Education Sciences, U.S. Department of Education.

Appendix A

Tables of Quality Metrics for Evaluating Alternative Approaches to Identifying Comparison Schools

Table A-1a – Match Quality Metrics for the "All Districts" Sample across Trials of the Estimated Magnet School Effect

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate				Mean of Magnet Coefficients				Root Mean Square Error of Magnet Coefficients			
		All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group
ELA	3-3	○ 6.5	◑ 7.3	● 7.8	◔ 7.2	◔ 0.0028	● 0.0060	○ 0.0014	◑ 0.0054	○ 0.0242	◑ 0.0268	◕ 0.0280	◔ 0.0249
	3-2	○ 6.3	◔ 7.5	◑ 7.7	● 8.2	○ 0.0002	◑ 0.0044	◔ 0.0006	● 0.0090	○ 0.0228	◑ 0.0256	● 0.0271	◔ 0.0239
	2-3	○ 7.6	◑ 8.2	● 9.0	◔ 7.7	◔ 0.0058	◑ 0.0064	● 0.0076	○ 0.0001	○ 0.0244	● 0.0300	◑ 0.0283	◔ 0.0259
	2-2	○ 6.5	◔ 7.9	● 8.7	◑ 8.5	◔ 0.0033	○ 0.0021	● 0.0063	◑ 0.0037	○ 0.0228	● 0.0288	◑ 0.0263	◔ 0.0246
MATH	3-3	○ 6.9	◔ 7.2	◑ 7.6	● 11.6	◔ 0.0038	○ 0.0035	◑ 0.0051	● 0.0127	◔ 0.0267	◑ 0.0299	● 0.0305	○ 0.0266
	3-2	○ 5.9	◑ 7.5	◔ 6.1	● 9.3	○ 0.0015	◑ 0.0053	◔ 0.0037	● 0.0110	○ 0.0250	◑ 0.0276	● 0.0284	◔ 0.0258
	2-3	○ 7.0	◑ 8.7	◔ 7.6	● 8.9	◔ 0.0048	◑ 0.0062	○ 0.0047	● 0.0079	◔ 0.0266	◑ 0.0290	● 0.0313	○ 0.0265
	2-2	○ 5.6	◑ 7.3	◔ 6.1	● 7.5	◑ 0.0025	○ 0.0007	◔ 0.0011	● 0.0073	○ 0.0244	◑ 0.0271	● 0.0286	◔ 0.0247
Average ELA/MATH	3-3	◔ 6.9	◔ 6.9	◑ 7.5	● 9.3	◑ 0.0033	○ 0.0019	◔ 0.0032	● 0.0100	○ 0.0235	◑ 0.0265	● 0.0270	◔ 0.0244
	3-2	◔ 7.0	○ 6.5	◑ 7.9	● 9.3	○ 0.0009	◑ 0.0039	◔ 0.0022	● 0.0111	○ 0.0216	◑ 0.0241	● 0.0250	◔ 0.0226
	2-3	◔ 7.5	● 8.3	◑ 8.1	◔ 7.5	◑ 0.0053	◔ 0.0038	● 0.0061	○ 0.0013	○ 0.0234	◑ 0.0256	● 0.0274	◔ 0.0249
	2-2	○ 6.9	◔ 7.6	● 7.9	◑ 7.7	◔ 0.0029	○ 0.0015	◑ 0.0037	● 0.0043	○ 0.0211	◑ 0.0231	● 0.0246	◔ 0.0228
Average Across Subject and Years of Pre/Post Data		6.7	7.6	7.7	8.6	0.0031	0.0038	0.0038	0.0070	0.0239	0.0270	0.0277	0.0248

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. See notes to Table 1 for explanation of circle symbols.

Table A-1b – Match Quality Metrics for the “Small Districts” Sample across Trials of the Estimated Magnet School Effect

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate				Mean of Magnet Coefficients				Root Mean Square Error of Magnet Coefficients			
		All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group
ELA	3-3	○ 6.5	◑ 7.3	● 7.8	◔ 7.2	◔ 0.0028	● 0.0060	○ 0.0014	◑ 0.0054	○ 0.0242	◑ 0.0268	◕ 0.0280	◔ 0.0249
	3-2	○ 6.3	◔ 7.5	◑ 7.7	● 8.2	○ 0.0002	◑ 0.0044	◔ 0.0006	● 0.0090	○ 0.0228	◑ 0.0256	● 0.0271	◔ 0.0239
	2-3	○ 7.6	◑ 8.2	● 9.0	◔ 7.7	◔ 0.0058	◑ 0.0064	● 0.0076	○ 0.0001	○ 0.0244	● 0.0300	◑ 0.0283	◔ 0.0259
	2-2	○ 6.5	◔ 7.9	● 8.7	◑ 8.5	◔ 0.0033	○ 0.0021	● 0.0063	◑ 0.0037	○ 0.0228	● 0.0288	◑ 0.0263	◔ 0.0246
MATH	3-3	○ 6.9	◔ 7.2	◑ 7.6	● 11.6	◔ 0.0038	○ 0.0035	◑ 0.0051	● 0.0127	◔ 0.0267	◑ 0.0299	● 0.0305	○ 0.0266
	3-2	○ 5.9	◑ 7.5	◔ 6.1	● 9.3	○ 0.0015	◑ 0.0053	◔ 0.0037	● 0.0110	○ 0.0250	◑ 0.0276	● 0.0284	◔ 0.0258
	2-3	○ 7.0	◑ 8.7	◔ 7.6	● 8.9	◔ 0.0048	◑ 0.0062	○ 0.0047	● 0.0079	◔ 0.0266	◑ 0.0290	● 0.0313	○ 0.0265
	2-2	○ 5.6	◑ 7.3	◔ 6.1	● 7.5	◑ 0.0025	○ 0.0007	◔ 0.0011	● 0.0073	○ 0.0244	◑ 0.0271	● 0.0286	◔ 0.0247
Average ELA/MATH	3-3	◔ 6.9	◔ 6.9	◑ 7.5	● 9.3	◑ 0.0033	○ 0.0019	◔ 0.0032	● 0.0100	○ 0.0235	◑ 0.0265	● 0.0270	◔ 0.0244
	3-2	◔ 7.0	○ 6.5	◑ 7.9	● 9.3	○ 0.0009	◑ 0.0039	◔ 0.0022	● 0.0111	○ 0.0216	◑ 0.0241	● 0.0250	◔ 0.0226
	2-3	◔ 7.5	● 8.3	◑ 8.1	◔ 7.5	◑ 0.0053	◔ 0.0038	● 0.0061	○ 0.0013	○ 0.0234	◑ 0.0256	● 0.0274	◔ 0.0249
	2-2	○ 6.9	◔ 7.6	● 7.9	◑ 7.7	◔ 0.0029	○ 0.0015	◑ 0.0037	● 0.0043	○ 0.0211	◑ 0.0231	● 0.0246	◔ 0.0228
Average Across Subject and Years of Pre/Post Data		6.7	7.6	7.7	8.6	0.0031	0.0038	0.0038	0.0070	0.0239	0.0270	0.0277	0.0248

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. See notes to Table 1 for explanation of circle symbols.

Table A-1c – Match Quality Metrics for the "Medium Districts" Sample across Trials of the Estimated Magnet School Effect

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate				Mean of Magnet Coefficients				Root Mean Square Error of Magnet Coefficients			
		All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group
ELA	3-3	◔ 9.8	○ 9.6	● 14	● 14	○ 0.0016	◔ 0.0042	◑ 0.0142	● 0.0165	◔ 0.0366	◔ 0.0393	◕ 0.0441	◔ 0.0362
	3-2	◔ 9.5	○ 9.3	◑ 12.6	● 14.4	○ 0.0001	◔ 0.0023	◑ 0.0114	● 0.0157	◔ 0.0334	◑ 0.0362	● 0.0415	○ 0.0333
	2-3	○ 8.2	◔ 11.4	◑ 12.8	● 14.9	◔ 0.0036	○ 0.0032	● 0.0109	◑ 0.0093	○ 0.036	● 0.0465	◑ 0.0441	◔ 0.0387
	2-2	○ 10.1	◔ 12.8	◑ 13.4	● 14.1	○ 0.0019	◔ 0.0071	● 0.0086	◑ 0.0073	○ 0.033	● 0.0437	◑ 0.0404	◔ 0.0359
MATH	3-3	○ 9.9	◔ 12.5	◑ 13.9	● 18.6	○ 0.0006	◔ 0.005	◑ 0.0155	● 0.0269	◔ 0.0426	◑ 0.048	● 0.0497	○ 0.0423
	3-2	○ 10.7	◔ 11.9	● 15.5	◑ 15	◔ 0.0034	○ 0.001	◑ 0.0169	● 0.0206	○ 0.0386	◑ 0.0439	● 0.0456	◔ 0.0407
	2-3	○ 10.3	◔ 11.7	◑ 14	● 16.4	◔ 0.0015	○ 0.0011	◑ 0.0093	● 0.021	○ 0.0417	◑ 0.045	● 0.0507	◔ 0.0437
	2-2	○ 9.6	◔ 11.8	● 14.1	◑ 13.1	◑ 0.0016	○ 0.0003	◔ 0.0014	● 0.0151	○ 0.0374	◑ 0.0418	● 0.0467	◔ 0.0412
Average ELA/MATH	3-3	○ 10	◔ 10.9	◑ 15.8	● 15.9	○ 0.0005	◔ 0.0071	◑ 0.0149	● 0.0168	○ 0.037	◑ 0.0421	● 0.0436	◔ 0.0388
	3-2	○ 10.8	◔ 12.7	● 14.3	◑ 13.5	○ 0.0018	◔ 0.0078	● 0.0141	◑ 0.013	○ 0.0325	◑ 0.0384	● 0.0391	◔ 0.0354
	2-3	○ 9.2	◔ 11.8	● 14.4	◑ 14	◔ 0.0026	○ 0.0017	◑ 0.0101	● 0.014	○ 0.0358	◑ 0.0395	● 0.0434	◔ 0.0377
	2-2	○ 9.4	◔ 11.5	● 13.4	◑ 13	○ 0.0001	◔ 0.0031	◑ 0.005	● 0.0122	○ 0.0311	◑ 0.0351	● 0.0384	◔ 0.0341
Average Across Subject and Years of Pre/Post Data		9.8	11.5	14.0	14.7	0.0016	0.0037	0.0110	0.0157	0.0363	0.0416	0.0439	0.0382

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. See notes to Table 1 for explanation of circle symbols.

Table A-1d – Match Quality Metrics for the “Large Districts” Sample across Trials of the Estimated Magnet School Effect

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate				Mean of Magnet Coefficients				Root Mean Square Error of Magnet Coefficients			
		All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group
ELA	3-3	○ 13.6	◑ 18.4	● 18.5	◔ 15.1	○ 0.0007	◔ 0.001	◑ 0.0066	● 0.0067	○ 0.0536	◕ 0.0648	◕ 0.0646	◔ 0.0564
	3-2	○ 14	● 18.3	◑ 16.1	◔ 15.4	○ 0	◔ 0.0029	◑ 0.0065	● 0.0067	○ 0.0519	◑ 0.0619	● 0.062	◔ 0.0567
	2-3	○ 14.3	● 19.1	◑ 16.7	◔ 16.3	◔ 0.0014	◑ 0.006	○ 0.0009	● 0.0091	○ 0.0514	● 0.0663	◑ 0.0614	◔ 0.0529
	2-2	○ 14.5	● 19	◑ 15.9	◔ 15.7	◔ 0.0009	◑ 0.007	○ 0.0007	● 0.0081	○ 0.0493	● 0.0646	◑ 0.0574	◔ 0.0523
MATH	3-3	○ 12.6	◑ 17.5	● 18.1	◔ 15.5	◑ 0.0037	● 0.0121	○ 0.0022	◔ 0.0029	○ 0.0589	◑ 0.0724	● 0.0729	◔ 0.0644
	3-2	○ 13.9	◔ 16.4	● 18.3	◑ 16.6	◑ 0.0016	● 0.0111	◔ 0.0011	○ 0.0007	○ 0.0577	● 0.0698	● 0.0698	◔ 0.0664
	2-3	○ 13.1	● 17.5	◑ 17.2	◔ 15.2	◑ 0.0039	● 0.0079	○ 0.0013	◔ 0.0023	○ 0.0571	● 0.0745	◑ 0.0692	◔ 0.0611
	2-2	○ 13.5	◑ 18	● 18.3	◔ 15.6	◑ 0.0021	● 0.0077	◔ 0.0007	○ 0.0003	○ 0.0557	● 0.0729	◑ 0.0687	◔ 0.0622
Average ELA/MATH	3-3	○ 13	● 18	● 18	◔ 14.1	◔ 0.0022	● 0.0058	◑ 0.0044	○ 0.0008	○ 0.0523	● 0.064	◑ 0.0638	◔ 0.0566
	3-2	○ 12.9	● 17.4	◔ 16.6	◑ 17.3	◔ 0.0008	● 0.0041	◑ 0.0038	○ 0.0007	○ 0.0503	◑ 0.0592	● 0.0601	◔ 0.0568
	2-3	○ 14.1	◑ 16.3	● 16.5	◔ 15.2	◑ 0.0027	◔ 0.0016	○ 0.0011	● 0.003	○ 0.0501	◑ 0.0602	● 0.0605	◔ 0.0521
	2-2	○ 11.8	● 18	◑ 17.8	◔ 15.7	◑ 0.0015	○ 0.0001	◔ 0.0007	● 0.0024	○ 0.0479	● 0.0578	◑ 0.0575	◔ 0.0513
Average Across Subject and Years of Pre/Post Data		13.4	17.8	17.3	15.6	0.0018	0.0056	0.0025	0.0036	0.0530	0.0657	0.0640	0.0574

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. See notes to Table 1 for explanation of circle symbols.

Table A-1e – Summary of Average Match Quality Metrics across Scenarios for "All Districts", "Small Districts", "Medium Districts" and "Large Districts" Sample across Trials of the Estimated Magnet School Effect

District Category	Magnet Coefficient Rejection Rate				Mean of Magnet Coefficients				Root Mean Square Error of Magnet Coefficients			
	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group	All Traditional Schools	Regression-Based Matching	Propensity Score	Synthetic Control Group
All Districts	○ 6.7	◔ 7.6	◑ 7.7	● 8.6	○ 0.0031	◔ 0.0038	◑ 0.0038	● 0.0070	○ 0.0239	◑ 0.0270	● 0.0277	◔ 0.0248
Small	◔ 11.0	◑ 11.7	○ 10.5	● 12.4	◑ 0.0065	● 0.0103	○ 0.0055	◔ 0.0064	○ 0.0372	◑ 0.0402	● 0.0424	◔ 0.0384
Medium	○ 9.8	◔ 11.5	◑ 14.0	● 14.7	○ 0.0016	◔ 0.0037	◑ 0.0110	● 0.0157	○ 0.0363	◑ 0.0416	● 0.0439	◔ 0.0382
Large	○ 13.4	● 17.8	◑ 17.3	◔ 15.6	○ 0.0018	● 0.0056	◔ 0.0025	◑ 0.0036	○ 0.0530	● 0.0657	◑ 0.0640	◔ 0.0574
Grand Average Across District Categories	10.2	12.2	12.4	12.8	0.0032	0.0058	0.0057	0.0082	0.0376	0.0436	0.0445	0.0397

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Each cell in the first three rows contains the average quality metric across all scenarios defined by combinations of test subject and availability of pre/post data for a given district category. Row four simply takes a grand average of the district-category specific values for the method in each column. See notes to Table 1 for explanation of circle symbols.

Table A-2a – Comparison of Match Quality Metrics from Simulation of <u>Regression-Based</u> Modeling of Magnet School Effects for the "All Districts" Sample Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores
ELA	3-3	7.3	7.5	0.0060	0.0076	0.0268	0.0273
	3-2	7.5	6.3	0.0044	0.0046	0.0256	0.0255
	2-3	8.2	7.3	0.0064	0.0027	0.0300	0.0291
	2-2	7.9	9.0	0.0021	0.0065	0.0288	0.0273
MATH	3-3	7.2	6.7	0.0035	0.0004	0.0299	0.0297
	3-2	7.5	7.0	0.0053	0.0047	0.0276	0.0275
	2-3	8.7	6.8	0.0062	0.0003	0.0290	0.0276
	2-2	7.3	6.0	0.0007	0.0022	0.0271	0.0255
Average ELA/MATH	3-3	6.9	6.5	0.0019	0.0043	0.0265	0.0253
	3-2	6.5	6.3	0.0039	0.0009	0.0241	0.0232
	2-3	8.3	7.6	0.0038	0.0026	0.0256	0.0252
	2-2	7.6	6.9	0.0015	0.0012	0.0231	0.0231
Average Across Subject and Years of Pre/Post Data		7.6	7.0	0.0038	0.0032	0.0270	0.0264

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric. Ties are signified by neither element of the pair being highlighted.

Table A-2b – Comparison of Match Quality Metrics from Simulation of <u>Regression-Based</u> Modeling of Magnet School Effects for Small-size Districts Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores
ELA	3-3	9.2	9.5	0.0090	0.0081	0.0401	0.0393
	3-2	9.6	9.7	0.0068	0.0004	0.0412	0.0367
	2-3	13.0	12.6	0.0181	0.0074	0.0454	0.0484
	2-2	11.1	11.5	0.0103	0.0020	0.0450	0.0464
MATH	3-3	10.1	11.0	0.0069	0.0005	0.0453	0.0391
	3-2	10.3	13.1	0.0087	0.0085	0.0434	0.0394
	2-3	16.7	13.8	0.0222	0.0127	0.0390	0.0384
	2-2	11.4	11.9	0.0084	0.0054	0.0360	0.0350
Average ELA/MATH	3-3	9.0	8.7	0.0052	0.0041	0.0371	0.0352
	3-2	8.1	7.9	0.0016	0.0030	0.0349	0.0320
	2-3	16.3	13.5	0.0160	0.0087	0.0397	0.0397
	2-2	16.0	13.0	0.0102	0.0050	0.0354	0.0365
Average Across Subject and Years of Pre/Post Data		11.7	11.4	0.0103	0.0055	0.0402	0.0388

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric. Ties are signified by neither element of the pair being highlighted.

Table A-2c – Comparison of Match Quality Metrics from Simulation of Regression-Based Modeling of Magnet School Effects for Medium-size Districts Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores
ELA	3-3	9.6	11.9	0.0042	0.0123	0.0393	0.0414
	3-2	9.3	11.5	0.0023	0.0125	0.0362	0.0388
	2-3	11.4	12.4	0.0032	0.0106	0.0465	0.0431
	2-2	12.8	14.8	0.0071	0.0121	0.0437	0.0395
MATH	3-3	12.5	11.6	0.0050	0.0005	0.0480	0.0501
	3-2	11.9	11.6	0.0010	0.0041	0.0439	0.0458
	2-3	11.7	11.8	0.0011	0.0099	0.0450	0.0442
	2-2	11.8	11.2	0.0003	0.0098	0.0418	0.0416
Average ELA/MATH	3-3	10.9	10.8	0.0071	0.0079	0.0421	0.0394
	3-2	12.7	11.0	0.0078	0.0033	0.0384	0.0351
	2-3	11.8	11.4	0.0017	0.0035	0.0395	0.0385
	2-2	11.5	11.6	0.0031	0.0025	0.0351	0.0343
Average Across Subject and Years of Pre/Post Data		11.5	11.8	0.0037	0.0074	0.0416	0.0410

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-2d – Comparison of Match Quality Metrics from Simulation of <u>Regression-Based</u> Modeling of Magnet School Effects for Large-size Districts Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores	Regression-Based Matching	Regression-Based Matching - using only lagged test scores
ELA	3-3	18.4	18.5	0.0010	0.0008	0.0648	0.0671
	3-2	18.3	18.9	0.0029	0.0030	0.0619	0.0633
	2-3	19.1	18.4	0.0060	0.0050	0.0663	0.0678
	2-2	19.0	19.3	0.0070	0.0031	0.0646	0.0659
MATH	3-3	17.5	17.1	0.0121	0.0069	0.0724	0.0728
	3-2	16.4	15.9	0.0111	0.0052	0.0698	0.0684
	2-3	17.5	16.0	0.0079	0.0006	0.0745	0.0691
	2-2	18.0	15.9	0.0077	0.0016	0.0729	0.0663
Average ELA/MATH	3-3	18.0	14.7	0.0058	0.0012	0.0640	0.0621
	3-2	17.4	16.5	0.0041	0.0004	0.0592	0.0592
	2-3	16.3	16.6	0.0016	0.0080	0.0602	0.0617
	2-2	18.0	17.1	0.0001	0.0057	0.0578	0.0578
Average Across Subject and Years of Pre/Post Data		17.8	17.1	0.0056	0.0035	0.0657	0.0651

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric. Ties are signified by neither element of the pair being highlighted.

Table A-3a – Comparison of Match Quality Metrics from Simulation of Synthetic Control Modeling of Magnet School Effects for the "All Districts" Sample Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores
ELA	3-3	7.2	8.8	0.0054	0.0077	0.0249	0.0256
	3-2	8.2	9.5	0.0090	0.0110	0.0239	0.0241
	2-3	7.7	8.0	0.0001	0.0016	0.0259	0.0260
	2-2	8.5	8.7	0.0037	0.0029	0.0246	0.0250
MATH	3-3	11.6	12.4	0.0127	0.0161	0.0266	0.0263
	3-2	9.3	11.7	0.0110	0.0155	0.0258	0.0253
	2-3	8.9	9.6	0.0079	0.0124	0.0265	0.0248
	2-2	7.5	9.8	0.0073	0.0122	0.0247	0.0232
Average ELA/MATH	3-3	9.3	9.6	0.0100	0.0089	0.0244	0.0243
	3-2	9.3	10.2	0.0111	0.0110	0.0226	0.0224
	2-3	7.5	8.3	0.0013	0.0029	0.0249	0.0244
	2-2	7.7	8.5	0.0043	0.0054	0.0228	0.0222
Average Across Subject and Years of Pre/Post Data		8.6	9.6	0.0070	0.0090	0.0248	0.0245

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-3b – Comparison of Match Quality Metrics from Simulation of Synthetic Control Modeling of Magnet School Effects for Small-size Districts Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores
ELA	3-3	10.2	11.4	0.0019	0.0054	0.0412	0.0433
	3-2	11.4	13.7	0.0115	0.0155	0.0396	0.0415
	2-3	12.3	12.4	0.0032	0.0042	0.0461	0.0465
	2-2	12.1	11.8	0.0079	0.0009	0.0438	0.0475
MATH	3-3	12.8	10.0	0.0037	0.0058	0.0356	0.0407
	3-2	11.6	12.6	0.0068	0.0124	0.0369	0.0416
	2-3	11.5	11.2	0.0011	0.0006	0.0371	0.0359
	2-2	14.3	12.4	0.0052	0.0059	0.0346	0.0341
Average ELA/MATH	3-3	13.3	10.5	0.0093	0.0054	0.0347	0.0370
	3-2	14.4	12.2	0.0158	0.0143	0.0321	0.0351
	2-3	13.1	12.2	0.0066	0.0042	0.0412	0.0419
	2-2	12.3	10.7	0.0035	0.0052	0.0378	0.0378
Average Across Subject and Years of Pre/Post Data		12.4	11.8	0.0064	0.0067	0.0384	0.0402

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric. Ties are signified by neither element of the pair being highlighted.

Table A-3c – Comparison of Match Quality Metrics from Simulation of <u>Synthetic Control</u> Modeling of Magnet School Effects for Medium-size Districts Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores
ELA	3-3	14.0	15.3	0.0165	0.0183	0.0362	0.0356
	3-2	14.4	15.4	0.0157	0.0183	0.0333	0.0330
	2-3	14.9	13.8	0.0093	0.0119	0.0387	0.0369
	2-2	14.1	14.1	0.0073	0.0118	0.0359	0.0339
MATH	3-3	18.6	14.5	0.0269	0.0200	0.0423	0.0397
	3-2	15.0	12.4	0.0206	0.0149	0.0407	0.0378
	2-3	16.4	14.8	0.0210	0.0205	0.0437	0.0396
	2-2	13.1	11.8	0.0151	0.0169	0.0412	0.0372
Average ELA/MATH	3-3	15.9	13.8	0.0168	0.0136	0.0388	0.0350
	3-2	13.5	13.4	0.0130	0.0117	0.0354	0.0321
	2-3	14.0	13.4	0.0140	0.0119	0.0377	0.0340
	2-2	13.0	12.1	0.0122	0.0104	0.0341	0.0308
Average Across Subject and Years of Pre/Post Data		14.7	13.7	0.0157	0.0150	0.0382	0.0355

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric. Ties are signified by neither element of the pair being highlighted.

Table A-3d – Comparison of Match Quality Metrics from Simulation of Synthetic Control Modeling of Magnet School Effects for Large-size Districts Predicted from *Pre-Intervention Test Scores and Student Demographics vs. Pre-Intervention Test Scores Only*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores	Synthetic Control Group	Synthetic Control Group - using only lagged test scores
ELA	3-3	15.1	15.2	0.0067	0.0041	0.0564	0.0553
	3-2	15.4	16.8	0.0067	0.0057	0.0567	0.0527
	2-3	16.3	17.2	0.0091	0.0042	0.0529	0.0524
	2-2	15.7	18.3	0.0081	0.0051	0.0523	0.0496
MATH	3-3	15.5	17.0	0.0029	0.0237	0.0644	0.0596
	3-2	16.6	17.3	0.0007	0.0181	0.0664	0.0580
	2-3	15.2	17.1	0.0023	0.0187	0.0611	0.0581
	2-2	15.6	15.8	0.0003	0.0134	0.0622	0.0564
Average ELA/MATH	3-3	14.1	14.6	0.0008	0.0077	0.0566	0.0544
	3-2	17.3	16.4	0.0007	0.0046	0.0568	0.0516
	2-3	15.2	15.9	0.0030	0.0023	0.0521	0.0519
	2-2	15.7	16.2	0.0024	0.0004	0.0513	0.0490
Average Across Subject and Years of Pre/Post Data		15.6	16.5	0.0036	0.0090	0.0574	0.0541

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-4a – Comparison of Match Quality Metrics from Simulation of Regression-Based Modeling of Magnet School Effects for the "All Districts" Sample When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math
ELA	3-3	7.3	6.3	0.0060	0.0040	0.0268	0.0277
	3-2	7.5	6.9	0.0044	0.0071	0.0256	0.0260
	2-3	8.2	7.3	0.0064	0.0010	0.0300	0.0278
	2-2	7.9	6.7	0.0021	0.0025	0.0288	0.0263
MATH	3-3	7.2	6.8	0.0035	0.0001	0.0299	0.0300
	3-2	7.5	6.0	0.0053	0.0008	0.0276	0.0275
	2-3	8.7	8.8	0.0062	0.0084	0.0290	0.0290
	2-2	7.3	7.8	0.0007	0.0055	0.0271	0.0264
Average Across Subject and Years of Pre/Post Data		7.7	7.1	0.0043	0.0037	0.0281	0.0276

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric. Ties are signified by neither element of the pair being highlighted.

Table A-4b – Comparison of Match Quality Metrics from Simulation of Regression-Based Modeling of Magnet School Effects for Small-size Districts When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math
ELA	3-3	9.2	8.9	0.0090	0.0006	0.0401	0.0417
	3-2	9.6	10.0	0.0068	0.0051	0.0412	0.0397
	2-3	13.0	13.9	0.0181	0.0074	0.0454	0.0453
	2-2	11.1	11.3	0.0103	0.0045	0.0450	0.0435
MATH	3-3	10.1	11.7	0.0069	0.0105	0.0453	0.0424
	3-2	10.3	11.0	0.0087	0.0081	0.0434	0.0400
	2-3	16.7	16.5	0.0222	0.0241	0.0390	0.0432
	2-2	11.4	15.0	0.0084	0.0160	0.0360	0.0387
Average Across Subject and Years of Pre/Post Data		11.4	12.3	0.0113	0.0095	0.0419	0.0418

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-4c – Comparison of Match Quality Metrics from Simulation of Regression-Based Modeling of Magnet School Effects for Medium Districts Sample When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math
ELA	3-3	9.6	12.2	0.0042	0.0050	0.0393	0.0422
	3-2	9.3	13.7	0.0023	0.0068	0.0362	0.0394
	2-3	11.4	10.8	0.0032	0.0075	0.0465	0.0418
	2-2	12.8	11.6	0.0071	0.0096	0.0437	0.0393
MATH	3-3	12.5	10.4	0.0050	0.0092	0.0480	0.0487
	3-2	11.9	11.1	0.0010	0.0088	0.0439	0.0459
	2-3	11.7	11.4	0.0011	0.0042	0.0450	0.0457
	2-2	11.8	11.8	0.0003	0.0034	0.0418	0.0420
Average Across Subject and Years of Pre/Post Data		11.4	11.6	0.0030	0.0068	0.0431	0.0431

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-4d – Comparison of Match Quality Metrics from Simulation of <u>Regression-Based</u> Modeling of Magnet School Effects for Large-size Districts When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math	Regression-Based Matching	Regression-Based Matching - Using Average of ELA and Math
ELA	3-3	18.4	18.1	0.0010	0.0080	0.0648	0.0654
	3-2	18.3	18.3	0.0029	0.0065	0.0619	0.0621
	2-3	19.1	17.7	0.0060	0.0002	0.0663	0.0622
	2-2	19.0	17.4	0.0070	0.0019	0.0646	0.0594
MATH	3-3	17.5	17.5	0.0121	0.0037	0.0724	0.0723
	3-2	16.4	17.7	0.0111	0.0016	0.0698	0.0675
	2-3	17.5	15.7	0.0079	0.0034	0.0745	0.0699
	2-2	18.0	17.3	0.0077	0.0022	0.0729	0.0695
Average Across Subject and Years of Pre/Post Data		18.0	17.5	0.0070	0.0034	0.0684	0.0660

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the regression-based matching method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-5a – Comparison of Match Quality Metrics from Simulation of Synthetic Control Modeling of Magnet School Effects for the "All Districts" Sample When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group - Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group - Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group - Using Average of ELA and Math
ELA	3-3	7.2	8.0	0.0054	0.0088	0.0249	0.0262
	3-2	8.2	9.4	0.0090	0.0115	0.0239	0.0251
	2-3	7.7	6.9	0.0001	0.0000	0.0259	0.0271
	2-2	8.5	9.0	0.0037	0.0046	0.0246	0.0258
MATH	3-3	11.6	10.1	0.0127	0.0110	0.0266	0.0283
	3-2	9.3	9.5	0.0110	0.0106	0.0258	0.0269
	2-3	8.9	8.1	0.0079	0.0027	0.0265	0.0281
	2-2	7.5	7.4	0.0073	0.0041	0.0247	0.0263
Average Across Subject and Years of Pre/Post Data		8.6	8.6	0.0071	0.0067	0.0254	0.0267

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-5b – Comparison of Match Quality Metrics from Simulation of Synthetic Control Modeling of Magnet School Effects for Small-size Districts When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math
ELA	3-3	10.2	13.3	0.0019	0.0114	0.0412	0.0423
	3-2	11.4	15.1	0.0115	0.0215	0.0396	0.0390
	2-3	12.3	13.7	0.0032	0.0106	0.0461	0.0509
	2-2	12.1	13.4	0.0079	0.0020	0.0438	0.0476
MATH	3-3	12.8	13.8	0.0037	0.0068	0.0356	0.0411
	3-2	11.6	13.4	0.0068	0.0097	0.0369	0.0408
	2-3	11.5	12.1	0.0011	0.0024	0.0371	0.0415
	2-2	14.3	12.2	0.0052	0.0052	0.0346	0.0402
Average Across Subject and Years of Pre/Post Data		12.0	13.4	0.0052	0.0087	0.0394	0.0429

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-5c – Comparison of Match Quality Metrics from Simulation of <u>Synthetic Control</u> Modeling of Magnet School Effects for Medium-size Districts When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math
ELA	3-3	14.0	14.8	0.0165	0.0155	0.0362	0.0399
	3-2	14.4	12.6	0.0157	0.0122	0.0333	0.0377
	2-3	14.9	16.2	0.0093	0.0175	0.0387	0.0390
	2-2	14.1	16.3	0.0073	0.0177	0.0359	0.0364
MATH	3-3	18.6	13.8	0.0269	0.0181	0.0423	0.0445
	3-2	15.0	13.5	0.0206	0.0137	0.0407	0.0425
	2-3	16.4	12.1	0.0210	0.0104	0.0437	0.0444
	2-2	13.1	13.1	0.0151	0.0067	0.0412	0.0429
Average Across Subject and Years of Pre/Post Data		15.1	14.1	0.0166	0.0140	0.0390	0.0409

Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric.

Table A-5d – Comparison of Match Quality Metrics from Simulation of <u>Synthetic Control</u> Modeling of Magnet School Effects for Large-size Districts When *Not Averaging vs. Averaging Pre-Intervention ELA and Mathematics Test Scores to Predict Outcomes*

Subject	Years of Pre/Post Data	Magnet Coefficient Rejection Rate		Mean of Magnet Coefficients		Root Mean Square Error of Magnet Coefficients	
		Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math	Synthetic Control Group	Synthetic Control Group Using Average of ELA and Math
ELA	3-3	15.1	15.5	0.0067	0.0065	0.0564	0.0576
	3-2	15.4	15.9	0.0067	0.0061	0.0567	0.0580
	2-3	16.3	16.3	0.0091	0.0071	0.0529	0.0520
	2-2	15.7	16.9	0.0081	0.0058	0.0523	0.0519
MATH	3-3	15.5	15.4	0.0029	0.0049	0.0644	0.0658
	3-2	16.6	16.7	0.0007	0.0046	0.0664	0.0660
	2-3	15.2	14.8	0.0023	0.0011	0.0611	0.0628
	2-2	15.6	16.6	0.0003	0.0011	0.0622	0.0615
Average Across Subject and Years of Pre/Post Data		15.7	16.0	0.0046	0.0047	0.0591	0.0594

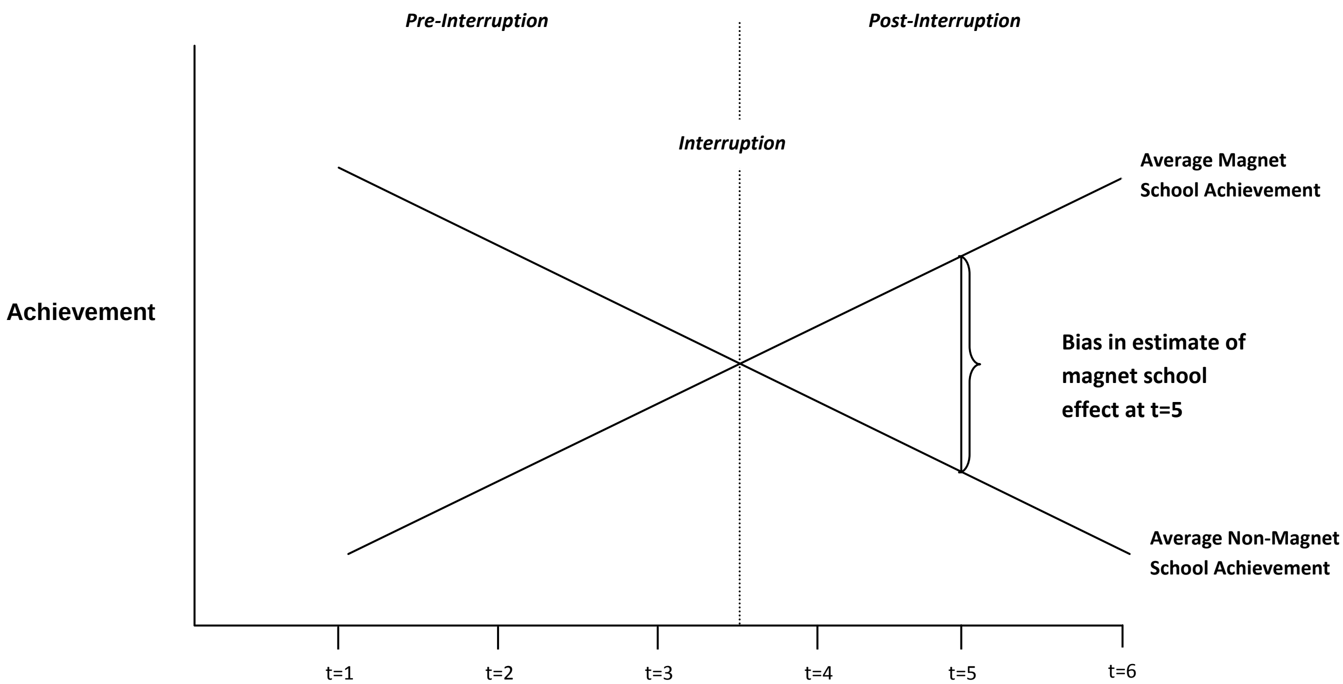
Note: Metrics include (a) the percentage of trials in which the null hypothesis of no difference in the three-year average post-interruption change in test scores between magnet and comparison schools is rejected at the 5% level of significance, (b) the absolute value of the mean of the magnet coefficients, and (c) the root mean square error of the magnet coefficients. Shaded cells indicate the synthetic control group method variant that proved better under each combination of subject, years of pre/post data and quality metric. Ties are signified by neither element of the pair being highlighted.

Appendix B – Example of Bias in CITS Estimator With Poor Matching Between Comparison and Treatment Groups

For a simple example of how deviating profiles can invalidate the CITS approach, consider the case shown in Figure B-1, in which the treatment and comparison group profiles are poorly matched: the former show achievement growth and the latter show achievement decline. In this example, the causal effect of treatment is zero in periods 4 through 6. It is true that the average treatment group test scores rise more quickly than do the average comparison school scores during years 4 through 6, but this was entirely due to the divergent pre-existing trends. Introducing treatment did not shift this group away from its pre-existing profile. By construction, the average overall profile prior to the interruption is zero because the individual treatment and comparison profiles cancel each other out. However, if we tested for an overall difference in test scores between the two groups in periods 4 to 6 while controlling for the average pre-interruption profile in test scores across the two schools, we would erroneously conclude that there is a large positive test-score difference in favor of the treatment group.[25] The figure shows the bias if the treatment effect was estimated for period 5. None of this gap in test scores can be causally attributed to the introduction of treatment. The "effect" would be entirely due to the researcher having chosen a comparison school with sharply divergent trends.

[25] A typical regression model would control for the average difference in scores between the two groups, which in this example is zero, and would add dummy variables for all but one period. Each of these dummies would have coefficient zero because, by construction in this example, the average overall test score follows a flat profile. When we then add three interactions between a variable indicating the treatment and dummies for years four, five, and six, these would have positive coefficients, capturing the differences such as the one for year four, shown in the diagram.

Figure B-1– Hypothetical Example of Bias in Comparative Interrupted Time Series Analysis (CITS) from Failure to Match on Pre-Interruption Trends



Appendix C– Technical Discussion of Regression-Based Matching

As mentioned above, one approach to matching appropriate comparison schools to treatment schools is to use a regression model to fit changes in the outcome variable using both the treatment and potential comparison observations prior to the policy reform, and to choose from the potential comparison group members the best matches for the group that later receives treatment. The following description closely follows the regression-based matching approach proposed in the NEMS analysis plan.

The available data includes school-level achievement outcomes on mathematics and ELA tests as well as selected student characteristics over the period 2001-02 to 2006-07. As we randomly assign schools to treatment status starting in the 2004-05 school year, there are up to three years of pre-interruption data with which to perform the matching. Using this data, a regression-based method is proposed to match magnet schools with comparison schools based on the predicted one-year change in achievement. The predictions are generated from equations estimated using lagged values of achievement, incidence of eligibility for federally subsidized free or reduced price lunch, and minority composition percentages:

$$(C.1-1)\quad \begin{aligned} &Y_{s,t=2003-04} - Y_{s,t=2002-03} \\ &\quad = \alpha + \beta_1 Y_{s,t=2001-02} + \beta_2 Y_{s,t=2002-03} + \beta_3 FRL_{s,t=2002-03} \\ &\quad + \beta_4 African\ American_{s,t=2002-03} + \beta_5 Hispanic_{s,t=2002-03} + \varepsilon_{s,t=2003-04} \end{aligned}$$

where *Y* is student achievement (mathematics or ELA), *FRL* is the percentage of students eligible for federally subsidized free or reduced price lunch, *African American* and *Hispanic* are the percentages of students in these minority groups, *ε* is a school-specific idiosyncratic error term, and the subscripts *s* and *t* denote individual schools and school years, respectively.[26] Two school-level regressions were run separately for each district in our sample using mathematics and ELA student achievement measures, respectively.

Next, we used the estimated equations to predict (forecast) the pre- to post-interruption achievement change from 2003-04 to 2004-05, the latter of which is the year immediately following magnet school program adoption:

$$(C.1-2)\quad \begin{aligned} &\widehat{Y_{s,t=2004-05} - Y_{s,t=2003-04}} \\ &\quad = \alpha + \hat{\beta}_1 Y_{s,t=2002-03} + \hat{\beta}_2 Y_{s,t=2003-04} + \hat{\beta}_3 FRL_{s,t=2003-04} \\ &\quad + \hat{\beta}_4 African\ American_{s,t=2003-04} + \hat{\beta}_5 Hispanic_{s,t=2003-04} \end{aligned}$$

[26] Note that only percent Hispanic and African American are included here. We believe these to be the most important minority groups to control for and feel this is prudent for many of the districts in our sample that have a small number of schools and, hence, offer a limited number of degrees of freedom to work with.

Within each district, the two non-magnet schools chosen were those for which the average forecasted change in achievement differed least from the forecasted change in achievement in the magnet school to which it is being matched.

In theory, one could control for two lags of demographics and additional demographic variables. The practical difficulty in doing this is that in small districts with fewer than a dozen elementary schools, even model (C.1 – 1) with its six coefficients will barely be identified. Conversely, one can imagine a simpler version of this equation that excludes the three demographic variables, on the grounds that it is plausible that these demographic characteristics do not contain information predictive of test scores beyond the information contained in the two lags of test scores already included on the right-hand side of the equation. Thus, we developed two separate sets of matches, one based on (C.1 – 1) and (C.1 – 2) and a second set based on similar models that drop *FRL, African American* and *Hispanic*.

In sum, we used the regression equations described above to predict the pre- to post-interruption test score change for each school, and thus to match to each magnet school the schools that have the closest predicted change in outcome in the given subject from pre- to post-interruption. The motivation behind this process is to identify comparison schools for each magnet that exhibit similar test score growth as magnet schools *had they not become magnets*. Forecasting the 2003-04 to 2004-05 test score changes with a model estimated using data through the 2003-04 school year exploits all of the information available up to the spring before the schools convert to magnet status.[27] There are two versions of this model estimated -- with and without school demographic variables as regressors.

In addition, as described in the text, we varied the number of years of data used pre- and post-treatment. The second and fourth scenarios in those variations, in which only two years of pre-treatment data are available, necessitates changes in the regression-based matching approach. Equations (C.1 – 1) and (C.1 – 2) above assumed the availability of three years of pre-treatment data. With only two test scores available prior to the year in which some schools are treated, we are forced to find comparison schools based on a match of test-score levels, rather than test-score gains. To see this, we show the analogues to (C.1 – 1) and (C.1 – 2) that we will have to use in the instance of only two years of pre-treatment data. The following assumes that when there are only two years of pre-treatment data (2002-03 and 2003-04) and that treatment, as before, begins in 2004-05:

$$(C.1-1') \quad Y_{s,t=2003-04} = \alpha + \beta_2 Y_{s,t=2002-03} + \beta_3 FRL_{s,t=2002-03} + \beta_4 African\ American_{s,t=2002-03} + \beta_5 Hispanic_{s,t=2002-03} + \varepsilon_{s,t=2003-04}$$

and based on the coefficients estimated from this model we then predict test scores in 2004-05:

[27] Note that the match is based on performance data prior to the 2004-2005 school year. It would be inappropriate to match based upon the 2004-05 test scores because magnet schools in that school year would have already initiated their magnet programs.

$$(C.1-2') \quad \widehat{Y_{s,t=2004-05}} = \alpha + \hat{\beta}_2 Y_{s,t=2003-04} + \hat{\beta}_3 FRL_{s,t=2003-04} + \hat{\beta}_4 African\ American_{s,t=2003-04} + \hat{\beta}_5 Hispanic_{s,t=2003-04}$$

Appendix D – Detailed Discussion of Synthetic Control Method

The synthetic control method was pioneered by Abadie and colleagues (Abadie and Gardeazabal, 2003 and extended by Abadie et al., 2007) and suggests a method for weighting observations from various members of the potential comparison group so as to create a "synthetic control group" that comes as close as possible to mimicking what the treated group would have experienced in the absence of receiving treatment. One then can evaluate the treatment effect by comparing post-interruption changes in the outcome variable(s) for the treated unit (school) against the weighted changes for the units (schools) in the synthetic control group. There are several variations to this approach presented in these two papers. Roughly speaking, the two main variants correspond to the two regression-based methods outlined above. The simpler method is based on past values of the outcome variable of interest, while the second also takes into account other variables that one thinks might have predictive power.

The more basic method matches based on all pre-interruption values of the outcome variable that are available, seeking to minimize the sum of squares of the differences in the outcome variable in the periods before treatment begins. Translating this method into the current context, suppose that there are J potential comparison schools. The synthetic control method chooses a vector of weights $W = (w_1,\ldots,w_J)'$ subject to $w_j \geq 0$, (j=1,2,…,J) and $\Sigma w_j=1$. Define Y_1 as the (3x1) matrix of test scores for a magnet school in the three years prior to treatment, and Y_0 as the (3xJ) matrix of corresponding test scores for the J potential comparison schools. In the simplest method, the optimal weights W^* are defined by the following:

$$(D.2-1) \qquad W^* = \arg\min_{w \in \Psi} \left((\mathbf{Y}_1 - \mathbf{Y}_0 \mathbf{W})'(\mathbf{Y}_1 - \mathbf{Y}_0 \mathbf{W}) \right)$$

where Ψ is the set of permissible weights.

The more involved strategy minimizes a weighted sum of squares of differences between the magnet school and the synthetic control group for test scores and other variables thought to be predictive of future test scores, which in our case would be the set of demographic variables listed in our description of the regression-based method above. The weights are chosen to minimize the sum of squared differences of the outcome variable (test scores) prior to treatment. Specifically, let X_1 and X_0 be a (Kx1) vector and a (KxJ) matrix consisting of the average pre-treatment values of K variables thought to be predictive of future test scores for the treatment school and pool of comparison schools, respectively. (For example we could set K to 4, and use the average over the three pre-treatment years of test scores and the three demographic variables listed earlier.) In this method, we choose the vector of weights W^* that solves the following (subject to the same constraints listed above):

(D.2 – 2) $$\min_{w \in \Psi} \left((X_1 - X_0 \mathbf{W})' V (X_1 - X_0 \mathbf{W}) \right)$$

where V is a diagonal matrix, the diagonal elements of which indicate the relative importance of the various variables in the X_1 and X_0 matrices, and Ψ is the set of permissible weights as described above. Each weighting matrix V will lead to a different set of weights W^*. The method chooses an optimal matrix V by comparing how well the optimal weights $W^*(V)$ for a given V match the pre-treatment test scores Y. The optimal V^* is thus given by

(D.2 – 3) $$V^* = \arg\min_{V \in \Omega} \left((\mathbf{Y}_1 - \mathbf{Y}_0 \mathbf{W}^*(V))' (\mathbf{Y}_1 - \mathbf{Y}_0 \mathbf{W}^*(V)) \right)$$

where Ω is the set of diagonal matrices with non-negative diagonal elements. In other words, the weights W^* used to create the synthetic control group are chosen first to minimize the weighted distance between a set of pre-treatment characteristics of the magnet school and the synthetic control school. The weights V^* that indicate the relative importance of test scores and the demographic variables are chosen such that the pre-treatment test scores of the magnet school and the synthetic control school are as close as possible.

In the synthetic control method, a weighted average of comparison schools is used as a single synthetic comparison school, so that each district *d* consists of the magnet and the synthetic comparison school. Our approach was not to use this weighted average as a single school but instead to use weighted least squares to estimate equation (1), with weights for the comparison schools being proportional to the synthetic control weights. To make this weighting scheme comparable to those implicit in our regression-based and propensity-score based matching methods, in which we chose the two closest matching comparison schools, we also set the sum of weights across the synthetic control schools equal to two.[28]

Appendix E – Other Matching Approaches Not Used

A fifth approach that deserves mention but is not included involves matching by hand. The researcher selects potential members of the comparison group to closely match trends in the treatment group before the policy reform. An example of this approach is the work by Card (1990) that studied the Mariel boatlift, in which a large inflow of Cuban refugees immigrated to the US, arriving on boats in Miami over a short period of time in 1980. He asks whether this large increase in the supply of unskilled workers affected the wages and employment of less-skilled natives in Miami. In this case the author implemented a difference-in-differences approach in which several southern U.S. cities were selected as

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[28] Because we found that using a sum of weights of one rejected the null hypothesis more often than when we used a sum of weights of two, in the main text we exclusively report results that use weighted least squares with a sum of weights of two for the control schools.

the comparison group for Miami. Here comparison units (cities) were selected with a view to match Miami's pre-existing employment trends.

Another approach is to choose matches that minimize the Mahalanobis distance between schools. Roughly speaking, this approach minimizes the weighted distance between all selected characteristics of schools.

During the feasibility analysis for the NEMS the research team similarly attempted to match magnet schools with other elementary schools by hand based on levels and changes of test scores in math and ELA and demographic characteristics. There it was concluded that in some cases it was difficult to match in an objective way by hand, and that simple but mechanical approaches such as minimizing the Mahalanobis distance between schools often led to matches with quite disparate changes in test scores. Given this initial experience, the research team concluded that alternative methods should be explored.

Appendix F – Further Details on Selection of States, Districts and Potential Schools, and Rationale for the Monte Carlo Approach

To be eligible for inclusion in the study, a state at a minimum had to have used the same math and ELA test over the six-year period from 2001-02 through 2006-07. Test scores had to be in a form that could be converted into Z-scores (scores with mean 0 and standard deviation 1 within year and grade based on statewide data). Based on our analysis of state test-score data, we included California, Florida and South Carolina in our study, using test scores from the 2001-02 through 2006-07 school years.[29]

We wanted our results to be relevant to studies of educational interventions in elementary schools nationwide, while probing for variations across district size in the relative effectiveness of the various methods. Accordingly, we used the sample of non-magnet elementary schools nationwide in the National Center for Education Statistics (NCES) Common Core of Data, and ranked districts by the number of non-magnet and non-charter elementary schools they contained. That is, we ranked them by their size as defined by the number of potential comparison schools. We then identified the districts in California, Florida, and South Carolina that included the number of schools at the 10th through 90th percentiles of district size nationwide. We also identified the district where the school at the 95th percentile is located, because our sense of the education intervention literature is that some of the largest districts in the country have hosted some of the most noteworthy educational interventions (e.g., Chicago, New York, San Diego among others). The first column in Table F-1 shows the number of elementary schools in the districts where the schools at the various percentiles are located.

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[29] South Carolina made minor changes to its ELA and math tests between 2001-02 and 2002-03. Our ELA of the state's documentation is that the test is sufficiently similar for the purpose of this study before and after these minor changes. We will thus use all six years of data, while controlling for year-specific effects.

Table F-1 – Nationwide Distributions of Numbers of Non-Magnet and Non-Charter Public Elementary Schools Across Districts in 2006-2007 (Weighted by Number of Elementary Schools)

Percentiles	Number of Elementary Schools Per District
10th	1
20th	2
30th	4
40th	5
50th	8
60th	11
70th	18
80th	30
90th	66
95th	133

Note: Calculations are based on Common Core of Data. Calculations based on 2003-2004 data produced similar results.

Schools at the 10th and 20th percentiles nationally (based on district size) would rarely if ever be candidates for inclusion in a CITS study: districts at the 10th percentile have only one elementary school, and so it is impossible to match treatment and comparison schools; districts at the 20th percentile have only two elementary schools per district, leaving only one comparison school which would not seem sufficient to make meaningful distinctions between the matching approaches. We therefore selected eight districts in California, ten in Florida, and six in South Carolina, in each state finding the districts that most closely matched the number of non-magnet schools listed in Table F-1 at the 30th through 90th and 95th percentiles.[30,31]

[30] In the case of South Carolina, where school districts tend to be quite small, we were not able to select a district matching the 90th percentile nationally, which had 66 schools. The same issue applied at the 95th percentile (133 schools). In turn, we used two districts from Florida that matched these numbers closely. Among the three states Florida has the highest number of large districts, because in that state school districts are contiguous with counties. The median elementary school in Florida resides in a district with 62 candidate comparison schools, which is between the 80th and 90th percentile nationally. Overall then, we have six districts from South Carolina, ten from Florida and eight from California.

[31] In the instance that two or more districts contain an identical number of non-magnet elementary schools, we randomly chose among them.

Within each district in the three selected states, we eliminated elementary schools from the analysis sample that: a) did not operate for the entire six-year period; or b) for which math and ELA scores were not available for all six years, or c) that were identified as magnet or charter schools based on the CCD and other state sources. Using the 55th percentile and the 85th percentile of the national distribution as the cutoff points distinguishing small and medium, and medium and large districts, respectively, the resulting sample contained 51 schools from nine small districts, 172 from nine medium districts and 496 from six large districts, for a total of 719 schools across 24 districts in the three states. (Thus, the average number of schools per district was about six, 19 and 83 for small, medium and large districts respectively.) Table F-2 below shows summary statistics for the variables used in our models.

Table F-2 – Means and Standard Deviations of Achievement Measures and Explanatory Variables for Sample of Schools in 24 Study Districts

Variable	Number of Observations	Mean	Standard Deviation	Minimum	Maximum
ELA Score	4,314	-0.002	0.377	-1.673	1.315
Math Score	4,313	-0.004	0.397	-1.454	1.329
Average of Math and ELA Scores_	4,313	-0.003	0.378	-1.476	1.246
Percentage of Students Eligible for Free and Reduced Price Lunch	4,314	55.662	25.571	0.000	99.761
Percentage of Students Who Are Hispanic	4,314	22.805	22.705	0.000	99.010
Percentage of Students Who Are African-American	4,314	27.163	27.762	0.000	100.000

Note: Sample includes data from 719 schools across 24 districts between the school years 2001-02 to 2006-07.

Two steps were followed to generate school-level ELA and math outcomes for the years 2001-02 to 2006-07 from the grade-specific mean-scale test scores for each school. First, within each grade the outcomes were *standardized* using the statewide student-level means and standard deviations (we will refer to these as "grade-specific standardized outcomes").[32] Next, school-level *pupil-weighted averages* of the grade-specific standardized outcomes were calculated for each school.

To select hypothetical treatment schools we chose a stratified random sample of traditional public schools – verified not to have adopted a magnet program – in which lower performing schools on state achievement tests had a greater probability of being selected. We considered them as receiving treatment beginning in the fourth year (2004-05) of the six-year sample. An immediate question is how many schools would have to be treated as if they were magnet schools to ensure that there would be sufficient statistical power to detect meaningful effects of treatment. Betts, Levin, Christenson and Eaton (2008) outline the proposed research plan for the ongoing national evaluation of MSAP-funded

[32] The reader will note that the ELA and math scores did not exhibit a unit variance (i.e., their standard deviations did not equal 1). This is because the grade-level mean scale scores for each school were standardized using statewide student-level standard deviations, which were as much as two time larger than the grade-level specific standard deviations.

schools. In this plan they estimate that for a sample of 21 magnet schools with data likely to be available for ELA, the Minimum Detectable Effect Size (MDES) was just under 0.17. Similarly, the likely sample of 23 magnet schools for math yielded a MDES of just under 0.17. Using a larger sample of 24 schools, according to power calculations we have done using the same method, yields an MDES of 0.15 and 0.16 for ELA and math, respectively.

Of particular interest was how often each method (using all non-treatment schools as comparisons, the regression-based method, the propensity score approach, and the synthetic control method) produced results suggesting that our randomly chosen magnet schools deviated from the achievement patterns of the comparison schools in the treatment years (2004-05 through 2006-07). If the experiment was performed just once, we would run the risk that a particular method could perform unusually well or badly simply due to chance. A standard way to circumvent this problem is repeatedly to draw samples of schools which we treat as magnet schools, and each time repeat the analysis. In this way we could characterize the distribution of outcomes across many trials.

Thus, as described earlier, we repeated the experiment 1,000 times. In each draw we randomly choose 24 schools from the three-state sample (one per district) and labeled these the hypothetical “magnet” (treatment) schools in 2004-05 through 2006-07. We had 719 elementary schools in the sample, thus roughly 97 percent of schools in any trial were available as potential comparison schools.[33] We then estimated the “effect” of hypothetical magnet school status for each method of selecting comparison schools (i.e., employing all possible comparison schools in district, regression-based matching, propensity score matching and the synthetic control method), using the same random draw of magnet schools.

One might question whether we really need 24 districts from three states. Couldn’t this all be done by selecting one or two districts in one or two states, and randomly selecting one or two schools? First, the number of schools that we select for the overall study of all district sizes is intended to approximate the NEMS study, and we imagine other CITS studies of educational interventions would need to have similar numbers of treated schools. Second, if we limited our study to just one district in one state, there could be concerns that something unobserved about that district or state was driving the results. Third, we select six to ten districts per state so as to capture the full range of district sizes observed in these states, with the exclusion of 10th and 20th percentile districts that are so small that they could never be used to perform a CITS (diff-in-diff) analysis within the same districts in the first place. Fourth, by selecting districts over a wide distribution of sizes, we can test for variations in the relative effectiveness of the matching methods across district size. Fifth, even with all of these precautions, if we perform a single random draw of 24 schools from our sample, and compare the

[33] Of course, schools in the smaller districts will each be picked multiple times as the magnet school, whereas in the 95th percentile district many schools will be picked only once. This does not defeat the purpose of the repeated random drawings, though, because in no case were the exact same combination of 24 schools will be drawn as magnet schools in two or more draws. Schools in the small districts were chosen on average 17.6 percent of the trials. Corresponding rates for schools in medium and large districts were 5.2 and 1.2 percent, respectively.

effectiveness of the various matching approaches, what is to say that our results did not derive from some unusual and atypical combination of the particular 24 schools that were chosen?

A common way to avoid this problem is perform a Monte Carlo simulation, that is, to repeat this matching exercise multiple times, each time drawing a new random sample of treated schools. Not only does this method reduce the risk of obtaining atypical results, but in addition it allows us to collect enough data to observe the distributions of the various statistics that we will calculate for each matching method. Because we select a wide variety of districts across three states, the approach will indeed draw a different sample of schools each time.

Appendix G. Conditioning on Average of ELA and Math in Small, Medium and Large Districts

Referring to Table 3 and Appendix Tables A-4a and A-5a, the main body of the report discussed the implications for all three evaluation metrics of conditioning on an average of pre-interruption ELA and math scores when using the regression-based and synthetic control approaches to identify comparison schools for the sample of all districts. Referring to Appendix Tables A-4b,A-4c, and A-4d for the regression-based approach in small, medium, and large districts, and Appendix Tables A-5b, A-5c, and A-5d for the synthetic control method in the different size districts, this appendix discusses the results of conditioning on the average of pre-interruption ELA and Math scores for samples of small, medium and large-size districts.

In small districts, our original regression-based matching method rejected closer to the nominal five percent level in six out of eight cases than the method that conditions on an average of ELA and math scores. The opposite occurred in medium and large district samples, though. In medium districts, the original method performed better by this metric in only two out of eight cases, respectively, and tied with the revised version that conditioned upon average scores in one other case. In the large-district sample, the original method performed better in one out of the eight cases, but tied with the revised method in two other cases. Based on our first metric, the implication is that a researcher who wants to perform a comparative interrupted time series analysis combined with regression-based matching on a similar sample of medium or large school districts might prefer the method that conditions upon the average of ELA and math scores over that which uses only the subject area that is being modeled in the comparative interrupted time series regression itself.

When we examined the two other metrics for the regression-based approaches, these patterns did not hold up as well. Our original regression-based matching method produced the lower coefficient mean in three of eight cases for the small districts, and the same pattern obtained when we examined root mean square error. Thus the advantage of the original method for small districts applies only if we focus on the percentage of cases in which the null hypothesis of no magnet effect was rejected. Conversely, in middle schools, where we found that the original method rejected more often than the revised matching method, the other two metrics provide a note of caution. The original method produced lower magnet school coefficient means in all eight cases for medium-sized districts, and lower root mean square error in six out of eight. For large districts, on the other hand, all three metrics suggest that the revised matching method that conditions upon the average of ELA and math scores performs better. The original method performed better than the revised "averaging" method in only two of eight cases when we examined the mean coefficient and root mean square error criteria.

Turning to comparisons between our original synthetic control method, which we matched based on the test score of interest (either ELA or math), and the revised synthetic control method that matched based on the average of ELA and math scores, results across the three metrics were slightly more consistent than they were for the regression-based matching method. These results appear in Appendix Tables A-5b, A-5c, and A-5d for small, medium, and large districts, respectively. Recall that for small districts, the original method outperformed in terms of the percentage of trials in which we rejected the null. Similarly, in seven out of eight cases the original method performed better in terms of our alternative metrics. For medium districts, the original method rejected closer to the five percent level in three out of eight cases, with one tie. The metric based on mean coefficients shows that the original method outperformed in only two out of eight cases. However, the original method outperformed in all eight cases when the metric considered was the root mean square error. In the case of large districts, the original synthetic control method slightly outperformed in terms of our main metric of percent significant, and performed about the same as the revised method in terms of the mean coefficient and the root mean square error.[34]

These results, taken as a whole, do not strongly endorse the use of the average of ELA and math scores in place of the actual test score being modeled. We have maintained that the most important selection metric is the percentage of cases in which the (true) hypothesis of no magnet effect is rejected. By this metric, cases in which a regression-based matching method might perform better in terms of rejection rates include analyses based on medium and large districts, with the opposite obtaining in small districts. For the synthetic control method, our original approach that conditions on the actual test score being modeled outperforms in the case of small and, to a lesser extent, large districts, but underperforms for medium districts.

[34] The number of cases (out of eight) in which our original synthetic control approach won was five (plus one tie), three and four for the metrics based on percent significant, mean coefficient, and root mean square error respectively.